

Metacognitive regulation of cognitive control to adjust choice accuracy in rhesus macaques

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Abstract

Making the best choice in one situation is critical, but it relies on limited cognitive capacities. Numerous animal species adjust the amount of cognitive resources allocated to a decision using a trade-off between speed and accuracy, where decision time corresponds to an investment of cognitive control, which scales with the expected increase in reward. In humans, recent studies indicate that control allocation is regulated not only based upon its potential benefits, but also upon its potential cost, through a metacognitive monitoring process. Is it the case in other primates? We addressed this issue in rhesus monkeys with a detailed analysis of the relation among choice accuracy, choice difficulty and response time in a two-choice task. In contrast to typical demonstrations of speed-accuracy trade-off, we found higher choice accuracy in trials with faster response time, above and beyond choice difficulty. This relationship can be explained by a model of metacognitive decision monitoring where the cost of cognitive control is traded against confidence benefits. Thus, rhesus monkeys also seem to rely upon supervisory monitoring to modulate the amount of control exerted over their decision efficacy. This might drive future research on the evolution of cognitive and neurophysiological processes underlying decision making.

Introduction

Balancing costs and benefits is one of the key determinants of animal behavior¹. It is a central feature of decision making, which consists of selecting courses of action that maximize the ratio between benefits and cost^{2–6}. However, the psychological and neurophysiological mechanisms through which the brain integrates costs and benefits to compute a representation of the subjective value of options, and arbitrates among these options through a decision process, are far from being understood^{7–13}. Decision-making generally occurs in complex environments in which the exact values of candidate options may be difficult to determine, because of individuals' biases, uncertain consequences and individual's limited computational resources. Resulting from these limitations, noise in the choice can arise in the form of an absence of marked (reliable) preference between alternatives, unveiling decision difficulty^{5,14}.

Capturing the idea of a common mechanism linking difficulty and accuracy, the drift diffusion models (DDMs) simulate the decision process as a noisy accumulation of evidence over time until a critical level of evidence is

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reached, terminating the process^{15–19}. Combined with the ‘dot-motion’ task, which manipulates the level of sensory evidence to study the dynamics of perceptual decision making in monkeys, this framework also provided a critical insight into its neural mechanisms^{20–23}. Using DDMs, this line of research captured the relationship between decision accuracy and decision difficulty, which is entangled with deliberation time, with difficult decisions associated to both slower and less accurate choices. As such, this framework can help to conceptualize the link between deliberation time and decision accuracy, and therefore the efficacy of the decision making process.

In addition to the identification of mechanisms underlying changes in decision accuracy as a function of decision difficulty, many studies also described how their mechanism could be adjusted to optimize accuracy and/or reward rate over time^{16,17,24,25}. A key finding in this field is that individuals display a decrease in accuracy when instructed or incentivized for speed (e.g., time pressure), suggesting that investing processing time improves decision accuracy (« speed/accuracy trade-off », ^{16,26}). This phenomenon has been observed not only in humans but also in other species including macaques, bees and rats^{26–28}. It has been suggested that in situations without instructions, this speed-accuracy trade-off could be optimally determined by individuals (human and non-human primates) to maximize reward rate over time^{29,30}, even though the mechanisms underlying the adjustment of this trade-off were lacking.

In that frame, the increase in processing time is interpreted as an increase in cognitive resources invested in the decision^{22,31}. This can be directly related to the notion of cognitive control, which refers to a set of cognitive mechanisms enabling agents to achieve specific goals, rather than following habits or reflexes^{32,33}. In the case of choice, exerting control improves choice accuracy when subjects must overcome a counterproductive bias, as it is typically the case in the Stroop task^{32,34}. In line with the principle of speed/accuracy trade off, quantitative studies demonstrated that the amount of time (taken as a proxy of cognitive control) invested in a difficult decision scaled with the amount of rewards that it could provide^{35–39}.

But investing cognitive control in a decision is inherently costly^{40–42}. Under this view, it can be considered as a form of mental effort and people avoid it if unnecessary^{43–46}. Indeed, in humans and monkeys performing the dot motion task, when controlling for the effect of difficulty (which, as mentioned above, strongly increases RT and decreases accuracy), shorter RTs are associated with higher accuracy^{22,26,28}. This could be accounted for by additional mechanisms, where, on top of the accumulation of evidence over time, subjects or monkeys could display an emergency pressure or a cost in the processing time^{26,30,31}. Indeed, recent studies in humans indicate that the exertion of control in the choice does not depend solely upon its potential benefits (increased rewards), but also upon its cost^{18,22,47–49}. Thus, one classical issue for decision making consists of understanding how injecting cognitive control in the decision can maintain choice accuracy (reliability) when choice difficulty increases (i.e., when the value contrast between the options decreases). This is usually captured using the notion of speed-accuracy trade-off, where spending more time (interpreted in terms of cognitive control) can improve accuracy. But on top of this ‘first level’ regulation, another critical issue consists of understanding a ‘second level’ (supervisory) regulation of the investment of control itself, based both upon its benefit (in terms of gain in accuracy) and its cost (in terms of resources invested).

In choice situations, recent studies suggest that gaining choice confidence may be what incentivizes the investment of costly cognitive resources in most types of decision, e.g., economic^{18,48,49} or perceptual⁴⁷. Thus, in humans, the supervisory process that regulates the allocation of control to the decision involves a metacognitive process where the level of control allocated to the decision corresponds to a given trade-off between the cost of cognitive control and the benefits (in terms of choice accuracy and reward) which maximizes confidence.

Although such mechanisms received support only in humans, confidence-based models can make qualitative and quantitative predictions regarding the natures of overt decision behavior—such as choice accuracy and reaction time—that can be observed in any animal species⁴⁸. First, such models could be insightful because the behavioral patterns reflecting an adjustment of decision efficacy could be observed in numerous species, but the underlying cognitive mechanisms remain unclear. Given the relative proximity in cognitive and brain organization between humans and other primate species, we will focus on primates. Indeed, there is clear evidence that

various primate species can mobilize cognitive control^{50–52}. As discussed above, models of decision making developed in humans also imply a supervisory process (to exert control over the control), as well as the notion of confidence^{47,48}. In humans, these notions are strongly associated with the notion of metacognition⁵³. In particular, we will consider the ‘online Metacognitive Control of Decision (oMCD) model, recently developed in our team⁴⁸. Even if metacognitive processes are relatively well characterized in humans, they could also be found in other species^{54,55}. Indeed, behavioral studies clearly established that several monkey species display metacognitive abilities, being able to adjust their actions according to their expected performance^{56–60}. However, even if several non-human primate species could display both cognitive control and metacognition, it is unclear whether monkeys could regulate the allocation of control in value based decision making, based on the expected costs and benefits of control allocation, as captured by the oMCD model⁴⁸. Here, we evaluated this cognitive mechanism in macaques (*Macaca mulatta*) and assessed the extent to which it could account for the complex relation among choice difficulty, choice accuracy and response time. Critically, our goal here was not to assess whether rhesus macaques could, or not, display some forms of metacognition, since this has already been established using standard task-based approaches^{61–63}. Our goal was to capture the nature of cognitive mechanisms regulating decision making, and assess the validity of the oMCD model developed in humans, in which the term ‘metacognition’ refers to a specific confidence-based supervisory process, well identified computationally^{47,48}.

We trained three monkeys to perform a binary choice task involving a compromise between two option features: reward benefits and physical effort cost. Each option’s value was defined by the weighted sum of reward benefits and effort cost, as inferred from each animal’s choices in the task. Choice difficulty could be evaluated by measuring the difference in value between the two offered options (left vs right), such that the hardest choices corresponded to small value differences between the two options. Importantly, both option features were sampled independently from each other, creating a diversity of compromises, hence inducing decisions with variable difficulty levels. We then measured choice accuracy, defined as the consistency between individual choices and their underlying subjective values¹⁴. Based on the assumption that the efficacy of the decision process is susceptible to vary from trial to trial, we looked for systematic variations in choice accuracy. We expected these variations to be related to indices of choice difficulty, but also to potential proxies of cognitive control. We considered a measure of decision time (response time), as well as a measure of decision preparation (spontaneous eye fixation right before option presentation). Choice accuracy and response times were affected not only by the options value, but also by the congruence between animals’ biases and the side of the best option, in keeping with classical manifestations of cognitive control in humans. To capture the cognitive mechanism that regulates the allocation of cognitive control in the decision, we described the tripartite relationship between choice accuracy, response time and choice difficulty and compared our results with a drift diffusion model, as well as oMCD, a model based on confidence and cognitive effort monitoring⁴⁸.

Results

Monkeys performed a binary choice task, where each choice option is defined in terms of two conflicting option features (reward size and required force level; see Fig. 1). We examined two behavioral responses: overt choice preferences (left or right option choice) and eye saccades (towards either the left or right option).

Measures of decision making

Overt choice (left or right option) was the instrumental response in the task. Recall that the amount of force exerted on the chosen lever needed to be greater than the required force level in order for the monkey to succeed in the trial and obtain the reward associated with the chosen side. Figure 2A shows that all three monkeys displayed a greater success rate when the required force levels were lower. As shown in Fig. 2B, monkeys adjusted the amount of exerted force to the required force level, which confirms that monkeys did not exert more force

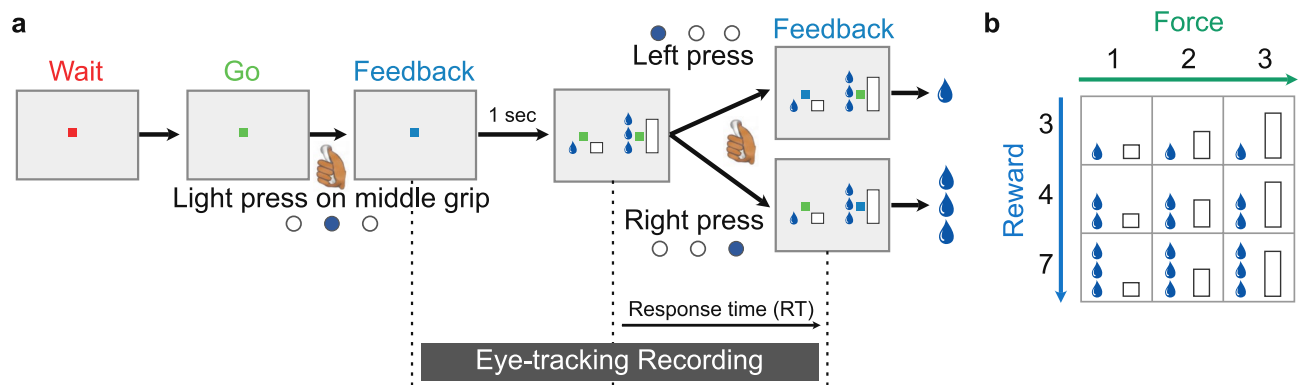


Fig. 1 Behavioral task design. **A.** Schematic representation of the sequence of screened events presented to the monkeys in the course of a trial. The inter-trial interval was set to 2 ± 0.1 s. Trials were automatically initiated with the display of a red dot in the center of the screen. The monkey could choose to engage in the task by touching the middle grip when the red dot turned green. Upon touch, the central dot turned blue and after 900 ± 200 ms delay period the two options were displayed. Each was characterized by a given level of required force (indicated by the size of the rectangle) and a given juice reward size (indicated by the number of drops). The monkey was free to evaluate both options for 2100 ms. If no option was chosen the trial was aborted and re-initialized with the exact same options for the next trial. To complete the trial monkeys had to exert the force corresponding to their choice on the corresponding lever (left or right). If the exerted force reached the required threshold, monkeys received the amount of reward corresponding to their choice. **B.** Representation of all the combination of force and reward for one option. Each combination was equiprobable. The three levels of force corresponded to three equidistant force thresholds to reach, the first one was a minimal force, and the third one was set for each monkey as the maximal force they were willing to exert in those conditions. The three levels of reward corresponded to 3, 4, and 7 drops of juice.

than necessary. This is evidence that monkeys perceived required force as costly. We next quantified how the required force level affected monkeys' preferences.

The choices of all three monkeys were clearly affected by both reward size and the required force level of the left and right options, as shown in Fig. 2C which presents the parameter estimates of a logistic regression predicting the choice from option features (see Eq. 1 in the Methods section). Supplementary Fig. 2A provides details on the probability to choose an option given force levels and reward sizes, and supplementary Fig. 2B shows the probability to choose the right option according to the difference of force or the difference of reward (right–left) between the two options. All these analyses show that monkeys avoided selecting options with high required force level but were more likely to select them if a higher reward was at stake. We then computed the theoretical subjective value of each option (left or right) by subtracting the weighted required force level from the weighted reward size on the corresponding side, where the weights were set from the logistic regression parameter estimates of Fig. 2C (see methods for further details). We then computed the difference of subjective value (hereafter ΔV) between the right and left options and quantified the rate of choosing the right option as a function of ΔV . Figure 2D shows that the choices of the monkeys are well explained by the difference of subjective value between the two options (for monkey R and J and to a lesser extent for monkey A). For all three monkeys, greater differences in subjective value between the two options led to greater probabilities of choosing the option with the highest subjective value. In conclusion, monkeys minimize the force they produce both in their choices and in action execution, but they are more willing to face the physical challenge when more reward is at stake. Moreover, monkeys rely upon the computation of a single decision variable (subjective value) sensitive to both reward and force levels to make their choices.

In order to get a supplementary measure of decision, we turned to eye-tracking analysis. Eye movements had no instrumental contingency whatsoever, in that they had no consequence on how monkeys were rewarded. Even though they were never trained to saccade in any particular manner, monkeys tended to fixate the central dot on the monitor before the onset of choice cues that would eventually occur on both sides of the central monitor.

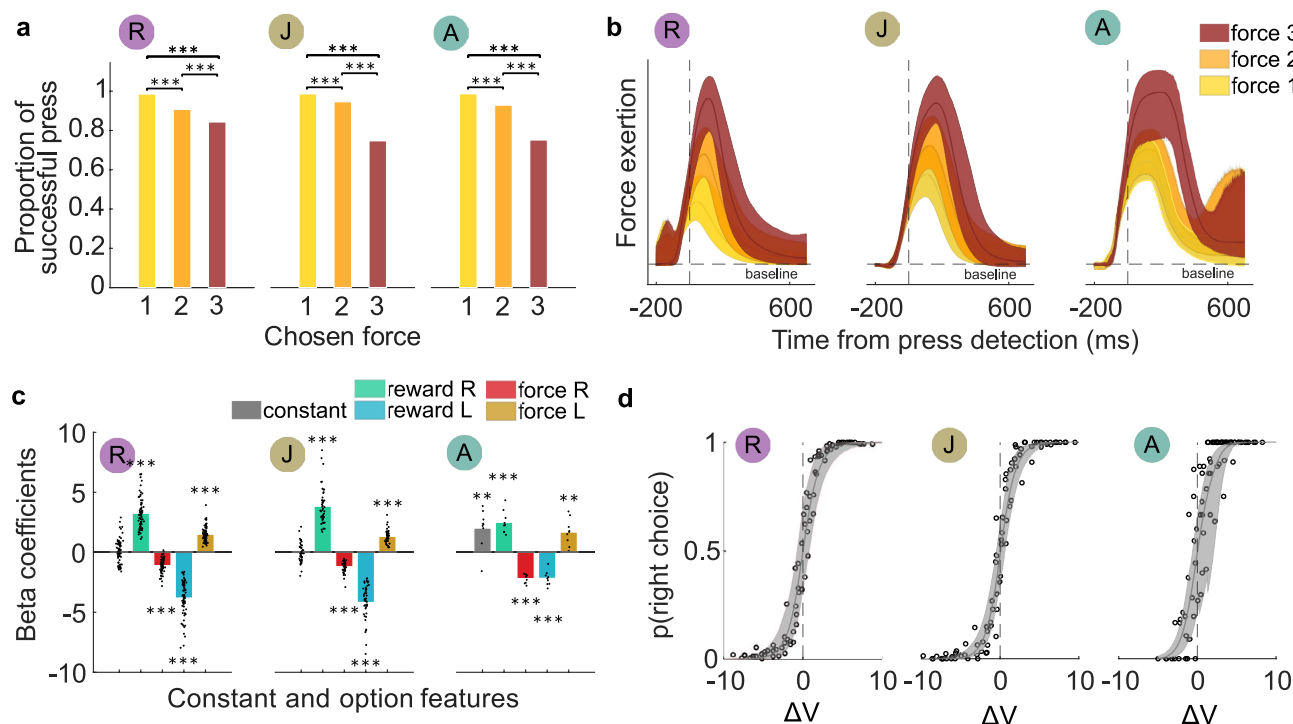


Fig. 2 Physical effort and cost-benefit trade-off. **A.** Bars representing the proportion of trials in which the required force threshold was successfully reached, for each of the three levels of required force. Proportions were compared with a z-test. **B.** Mean traces of force exerted by the monkeys on the lateral grips for each level of required force. Central line represents the mean and shaded area represents the standard deviation. **C.** Bars representing the mean value of beta coefficients of a logistic regression predicting the choice (right=1, left=0) from task parameters, performed on each behavioral session. The dots represent the value of the coefficients for each session. The formula of the regression is the following: $p(\text{right choice}) = \text{logit}(\beta_{\text{cst}} + \beta_{\text{rewR}} \cdot \text{rewR} + \beta_{\text{forR}} \cdot \text{forR} + \beta_{\text{rewL}} \cdot \text{rewL} + \beta_{\text{forL}} \cdot \text{forL})$, with cst: constant, rew: reward, for: force, R/L right vs left option. The value of the constant (cst) captures a potential side bias, with negative and positive values capturing a left vs right bias, respectively. We assessed the significance of the effect of each parameter on the choice with a t-test. **D.** The sigmoid is the result of a logistic regression estimating the probability to choose the right option depending on the subjective value contrast (ΔV) between the right and the left option. ΔV was calculated using the beta weights estimated with the regression presented in C. The grey shaded area represents the standard deviation of this fit across sessions. Each dot represents the proportion of right choice for one of the 81 possible compromises generated by the task and reported to its subjective value. * $p < 5 \cdot 10^{-2}$; ** $p < 5 \cdot 10^{-3}$; *** $p < 5 \cdot 10^{-4}$.

After cue onset, monkeys generally moved their gaze toward the side of the option (left or right) that they would eventually choose. This effect was visible when averaged across all trials for all three monkeys (Fig. 3A) and at the trial level for monkey R and J (Fig. 3B). We defined the eye choice and response time (eRT) as respectively the direction (right or left) and timing of the first saccade occurring at least 100 ms after cues onset (see methods for details). The response time of the (instrumental) hand response was defined as hRT. The average hRT was 520 ms for monkey R, 860 ms for monkey J and 470 ms for monkey A (Supp. Figure 2C) while the average eRT was 159 ms for monkey R, 295 ms for monkey J and 276 ms for monkey A (Supp. Figure 2E). Distribution of response time of both hRT and eRT can be found on Supplementary Fig. 2D and 2F respectively.

To better characterize eye choices, we measured how they were influenced by option features, with a logistic regression. As shown in Fig. 3C, the direction of the first saccade was clearly modulated by both reward size and required force level for monkey R and to a lesser extent for monkey J. In monkey A, however, the direction of the first saccade did not show any significant effect of either option feature. To further examine the relation between eye choices and the value comparison that drives subsequent hand choices, we ran a logistic regression of saccade directions onto subjective value difference (ΔV). As shown on Fig. 3D, the direction of the first saccade

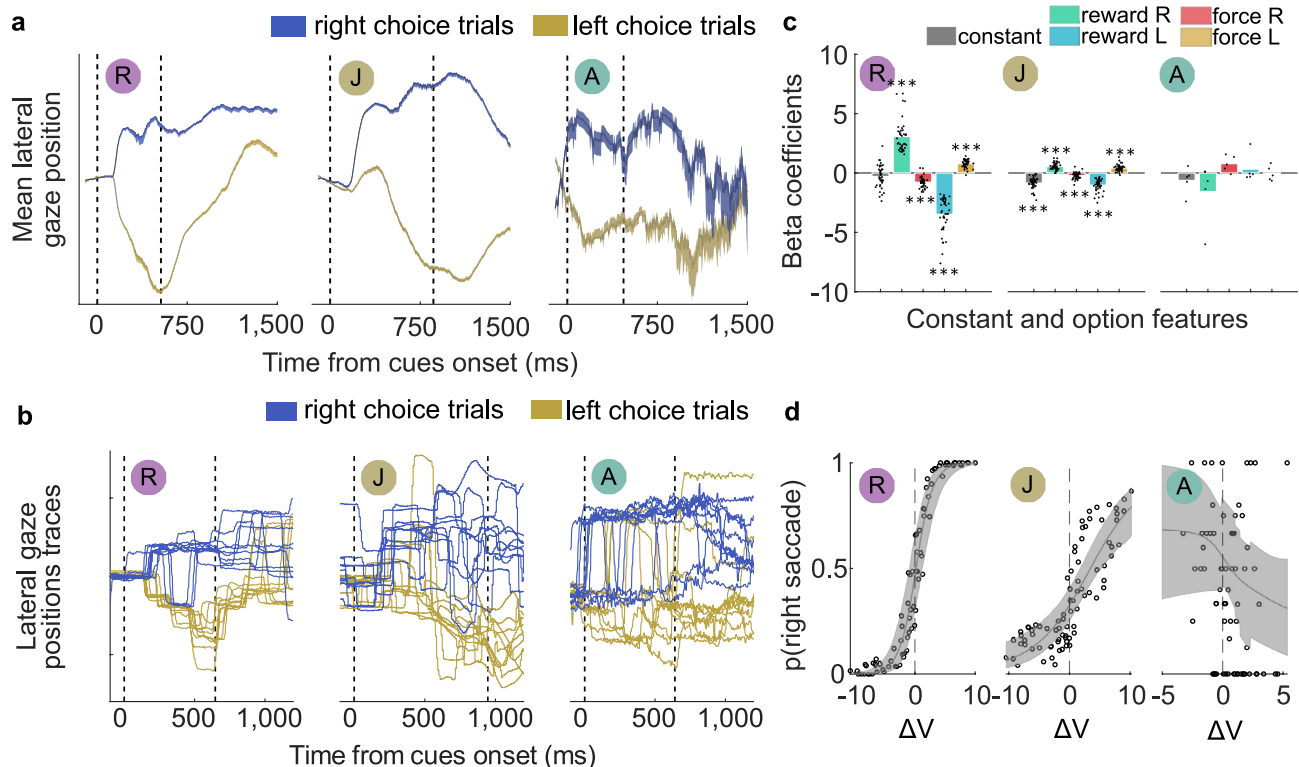


Fig. 3 Relation between choice and eye-tracking measures. **(A)** Mean trace of gaze position (on the lateral axis, i.e., left-right) during the course of trials from cue onset. Each of the two traces represents the mean trace according to the chosen side (left or right). The two dots lines represent the onset of cues representing options for the one at $t=0$ and the mean hand response time to reach lateral grip for the other one. **(B)** Traces of the position of the gaze of 20 randomly selected trials on a representative session. The color indicates the direction of the subsequent choice. **(C)** Same representation as Fig. 2C, for direction of the first saccade (eye choice). Bars represent the mean value of beta coefficients of a logistic regression predicting the choice (right=1, left=0) from task parameters, performed on each behavioral session. The dots represent the value of the coefficients for each session. We assessed the significance of the effect of each parameter on the choice with a t-test. **(D)** The sigmoid was obtained using a logistic regression estimating the probability that the first saccade was directed to the right depending on ΔV . Subjective difference of value was calculated using the hand choices (Fig. 2C). The grey shaded area represents the standard deviation of this fit across sessions. Each dot represents the proportion of right choices for one of the 81 possible compromises generated by the task and reported to its subjective value. * $p < 5 \cdot 10^{-2}$; ** $p < 5 \cdot 10^{-3}$; *** $p < 5 \cdot 10^{-4}$.

is significantly explained by ΔV for monkey R and J, with high $|\Delta V|$ leading to an almost certain probability to choose (with the eyes) the option with the highest subjective value, similar to what is observed for subsequent hand choice. Monkey A did not display this pattern at all. Thus, for two of the three monkeys (monkey R and J), the difference in value between the two options resulted in a relatively rapid saccade towards the most valuable option (based on subjective preference). To further examine the possibility that eye movements and hand movements were two behavioral manifestations of a common underlying cognitive process involving the comparison of option values, we measured the strength of the relation between eye and hand choice. First, the sensitivity to ΔV is greater for hand choices than for eye choices (Fig. 4A). This is expected, given that ΔV is estimated based on hand choices. Second, as shown on Fig. 4B, the hand and eye choices showed distinct levels of congruence across the three monkeys (93% for monkey R, 76% for monkey J and 50% for monkey A). When inspecting variations of eye and hand choice sensitivities to ΔV across sessions, we found that their correlation was significant only for monkey R (Fig. 4C), which is therefore likely to express overt eye and hand choices from the same process. A trend can be observed for monkey J, in line with the weaker effect of ΔV on eye choices (Fig. 3D). The absence of correlation for monkey A is expected, given the lack of sensitivity of eye saccades to options value.

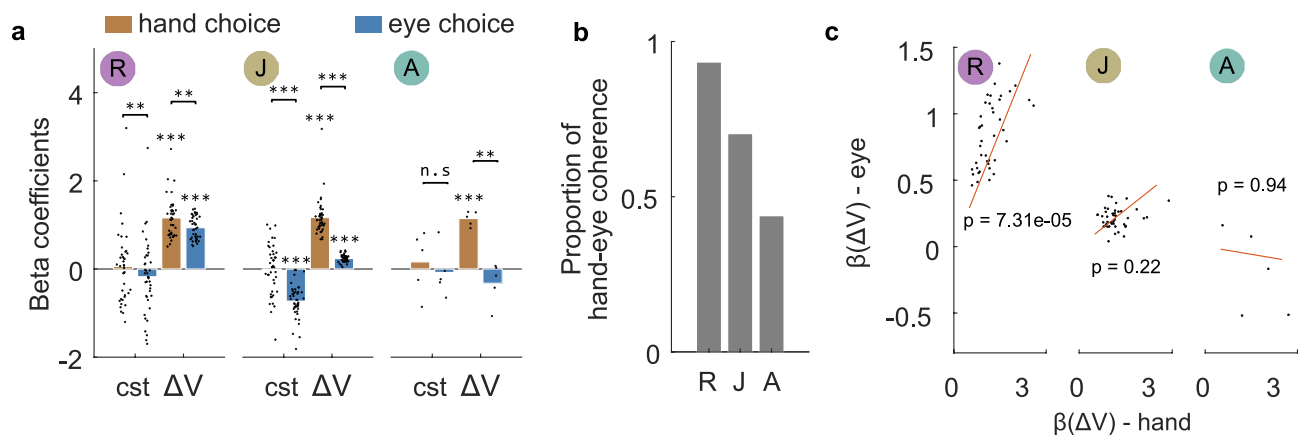


Fig. 4 Comparison of sensitivity of eye and hand choice to ΔV . **(A)** Comparison of the mean beta coefficients of two logistic regression predicting respectively eye and hand choice from the subjective difference of value between options (ΔV). The formula of the regression (predicting eye or hand choice) is: $p(\text{right choice}) = \text{logit}(\beta_{cst} + \beta \Delta V \cdot \Delta V)$. The bars represent the mean beta coefficients of the regressions performed on each session. Mean beta coefficients for the regression predicting hand choice are represented with orange bars. Mean beta coefficients predicting eye choice are represented with blue bars. The dots represent the betas of the regression for each behavioral session. We assessed the significance of the effect of each parameter on the choice with a t-test. We compared the mean beta coefficients from the two regressions with a t-test. **(B)** Proportion of trials where the side indicated by the eye and by the hand choice were the same (coherence), for each monkey. **(C)** Correlation between the sensitivity of the eye and the hand choice to ΔV , assessed from the logistic regression presented A. The position of each dot is determined by the estimation of $\beta(\Delta V)$ from the regression performed on hand choice (abscissa) and eye choice (ordinate) of the same behavioral session. The red line represents the slope of a linear regression between the two betas. * $p < 5.10^{-2}$; ** $p < 5.10^{-3}$; *** $p < 5.10^{-4}$.

Thus, several behavioral measures indicate that monkeys compute the value of each of the two options based on their expected reward benefits and effort costs, and select the best option accordingly. We next evaluated the extent to which choices could be modulated by cognitive control in this task, by exploring the relation between choice accuracy, choice difficulty and response time.

Decision accuracy: allocating control to overcome biases?

Selecting the best option can become more challenging, and therefore require more cognitive control, when action selection must overcome a bias. To examine the potential implication of cognitive control in the monkeys' decisions, we considered two potential response biases that could compete with the selection of the best option: a side bias and a repetition bias. We defined the side bias as a tendency to select the option on a given side, irrespectively of the values of the options presented on each side. We defined the repetition bias as a tendency to select the option located on the same side as the one selected on the previous trial, irrespectively of the values of the options presented on each side. We quantified these two biases using a logistic regression on choices (see “Materials and methods”). There was a significant side bias for all three monkeys (two on the left, one on the right, $p < 0.01$). There was a significant repetition bias for two of the three monkeys (J and R, $p < 0.01$), but the repetition bias did not reach significance for monkey A ($\beta_{\text{prev. choice}} = 0.3$, $p = 0.15$).

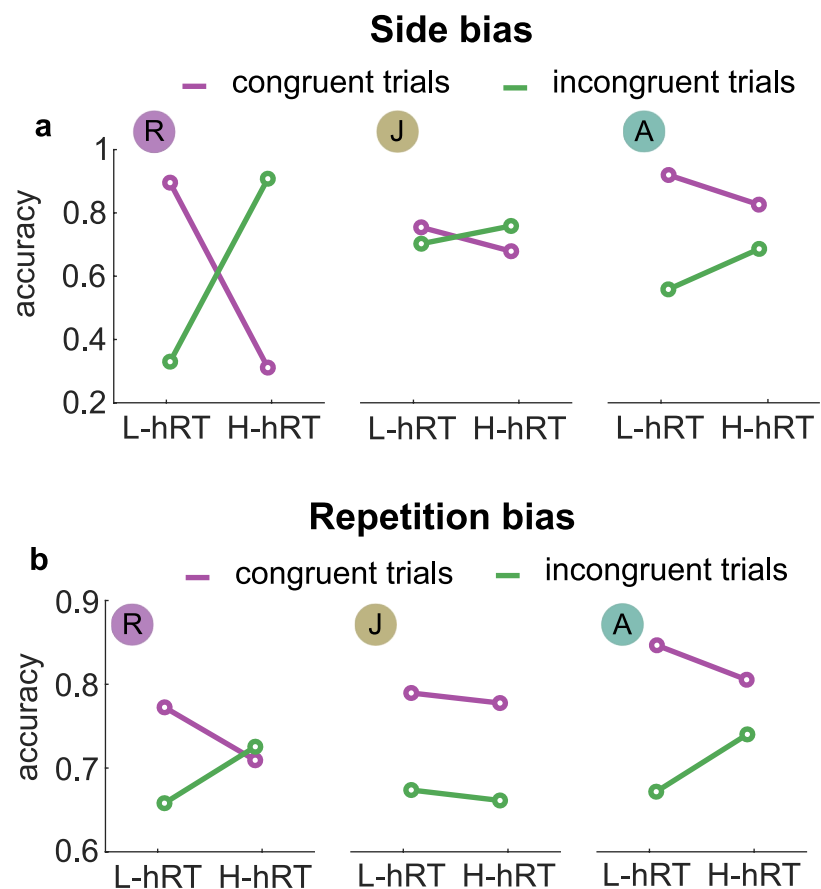
For each of these biases, we next assessed the extent to which choice accuracy (consistency between choices and option values) was affected by the congruence between the bias and the value contrast. Our underlying assumption was that in trials where the two response modes (following the bias and following the highest value) were incongruent, choices should be more difficult, and thus less accurate. Choice accuracy was defined as a binary variable taking the value of 1 when the animal chose the best option and 0 if the animal selected the other one. Here, the best option was identified by computing ΔV_{val} , which is based only on monkey's sensitivity to reward and force without taking into account the side bias, and differ from previously introduced ΔV in this

regard (see methods for details). Congruence was defined as a binary variable taking the value of 1 when the side of the bias was the same as the side of the best option, and 0 when the bias and the best option were on opposite sides. Finally, since choices are usually more challenging as the difference in value between the two options decreases, we included the absolute difference in subjective values ($|\Delta\text{Val}|$) in the model. As expected, $|\Delta\text{Val}|$ had a significant positive influence on choice accuracy in all monkeys ($\beta_{\text{adv}} > 0$; $p < 0.05$), which confirms that it contributes to determine how easy it is for the monkeys to identify the best of the two options.

For the side bias, only monkey A displayed a significant decrease in choice accuracy in incongruent compared to congruent trials ($\beta_{\text{congru}} = 0.7$, $p = 3.2 \cdot 10^{-10}$). The effect failed to reach significance for monkeys J and R. For the repetition bias, all three monkeys displayed a significant decrease in choice accuracy in incongruent compared to congruent trials ($\beta_{\text{congru}} = 0.29$; 0.34 and 0.12 for J, A and R, respectively, all p values < 0.001 .) Together, these data indicate that all three monkeys had at least one systematic response bias, which could favor (in congruent trials) or antagonize (in incongruent trials) the selection of the best option.

To evaluate the potential mobilization of cognitive control (using response time as a proxy) to counteract this bias, we examined the modulation of choice accuracy by the interaction between hRT and congruence, still controlling for value-based choice difficulty (using $|\Delta\text{Val}|$). For all monkeys in the case of the side bias and for monkeys A and R in the case of the repetition bias, congruence had a significant ($p < 0.05$) influence on hRT, in that monkeys were significantly faster in congruent compared to incongruent trials. The influence of hRT and congruence on choice accuracy are shown on Fig. 5. Note that we only used median split values of hRT for display purposes in Fig. 5, but the analyses were conducted on original data. For the side bias (Fig. 5A), all three monkeys displayed a significant interaction between hRT and congruence ($\beta_{\text{inter}} < 0$, p value < 0.01). As shown on Fig. 5A, hRT had a positive relation with choice accuracy in incongruent trials, but it decreased with choice accuracy in congruent trials. For the repetition bias (Fig. 5B), only monkey R displayed a significant interaction

Fig. 5 Influence of reaction time on accuracy in congruent vs incongruent conditions. Hand reaction times (hRT) were grouped into two categories, low (L) vs high (H), using a median split procedure. Note that we only used that procedure for display purpose. Congruence was defined at each trial as the coherence between the side determined by the bias and the side where the monkey's favorite option was located, based on its subjective preferences. The accuracy was defined as the ability to select the best option, based on individual subjective preferences. **(A)** Each panel corresponds to an individual monkey, indicated by a letter (R, J or A). For the side bias, the interaction between congruence and hRT has a significant influence on accuracy for all 3 monkeys ($p < 0.05$). In incongruent trials, accuracy increases with RT whereas it decreases with RT in congruent trials. **(B)** For the repetition bias, the interaction between hRT and congruence is significant only for monkey R, even if the same trend can be observed for monkey A.



between hRT and congruence ($\beta_{\text{inter}} < 0$, p value < 0.01). In monkey J, even though hRT seems to affect choice accuracy negatively regardless of congruence, this effect failed to reach significance ($\beta_{\text{RT}} = 0.7 \cdot 10^{-4}$, $p = 0.7$). No significant effect of hRT and no interaction reached significance in monkey A either ($p > 0.1$).

In short, for all three monkeys facing the side bias and for one of them facing the repetition bias, we observed a significant difference between the influence of hRT on accuracy in congruent vs incongruent trials. In incongruent trials, when monkeys need to overcome a bias in order to choose the best of the 2 options, the influence of hRT on performance is positive, whereas this influence is negative in congruent trials, which are less cognitively challenging. Even though this data could be interpreted in terms of cognitive control, it remains difficult to evaluate quantitatively the regulation of control as a function of its costs and benefits. Thus, we explored the influence of another, continuous measure of choice difficulty on choice accuracy and reaction times.

Decision accuracy: relation between value contrast and reaction time

As noted above, decision accuracy decreases when the difference in value between the two options decreases. Thus, the absolute difference in subjective values ($|\Delta V|$) constitutes a good proxy for how easy it is for the animals to identify the (subjectively) best option, based on the signed ΔV . As they seem to be able to do when facing biases, individuals may increase the level of cognitive control invested in the decision to compensate for choice difficulty and maintain a relatively good discrimination accuracy, which should be associated with a change in response time. Thus, we examined the three-way relationship between choice difficulty, choice accuracy and response time as a signature of cognitive control over the decision process. We used both hand and eye response time (hRT and eRT) to evaluate this relation.

We used linear regressions to explore the relation between choice difficulty (using $|\Delta V|$ as a measure of choice easiness) and response times (hRT and eRT). As shown in Fig. 6A and B, $|\Delta V|$ had a significant negative effect on at least one response time measure per monkey, i.e., a greater $|\Delta V|$ is typically associated with shorter response times. None of the measures of response time were positively influenced by $|\Delta V|$. For monkey R, both hRT and eRT were sensitive to $|\Delta V|$, with a lower regression coefficient for hRT than for eRT (hRT: $\beta_{|\Delta V|} = -0.09$, $p < 10e-05$; and eRT: $\beta_{|\Delta V|} = -0.38$, $p < 10e-21$). Note that the lower sensitivity of hRT to $|\Delta V|$ is due to the high influence of the chosen side on hRT, which disturbs the structure of the relation between hRT and accuracy, hence blunting its correlation with $|\Delta V|$ (see Fig. 6C, top, monkey R). For monkey J, hRT was sensitive to $|\Delta V|$ ($\beta_{|\Delta V|} = -0.23$, $p < 10e-18$) but not eRT ($\beta_{|\Delta V|} = 0.01$, $p = 0.29$). For monkey A, only hRT was sensitive to $|\Delta V|$ ($\beta_{|\Delta V|} = -0.32$, $p < 10e-06$, Fig. 6C). In line with earlier analysis, showing a lack of relation between eye movements and the decision in this monkey, eRT did not show any relation with $|\Delta V|$ ($\beta_{|\Delta V|} = -0.02$, $p = 0.83$). As shown on Fig. 6C, top for hRT and bottom for eRT, the negative relation between RT and $|\Delta V|$ was stable across the whole difficulty range. In short, in this task, trials with more difficult choices (smaller $|\Delta V|$) were associated with longer response times.

To explore the possibility that monkeys could mobilize cognitive control in order to maintain correct choice accuracy in the face of challenging choices (smaller $|\Delta V|$), we examined the relation between response time and choice accuracy (consistency between choice and options values) *over and above the influence of difficulty level*. For each monkey, we split the trials into two groups of matched difficulty levels, but distinct response times ('fast' and 'slow', see "Materials and methods" for further details). Additionally, to remove the potential confounding effects of choice side (left vs right) on RTs, we z-scored response times within each chosen side (cf. Material and Methods). Note that this transformation does not affect the relationship between response time and $|\Delta V|$ (Supp. Figure 3). We examined the consistency between choice and value in slow vs fast trials. For each measure (eye or hand), we performed a logistic regression to look at the relation between the response time (hRT or eRT) and the side indicated by the same measure (chosen side or direction of first saccade). As shown on Fig. 7A and C, the relation between choices and ΔV generally appears steeper in fast trials, with a significant interaction between ΔV and the 'slow/fast' categorization (Fig. 7B and D). None of the measures showed the opposite effect (i.e., slower reaction time associated with greater consistency). For monkey R, the effect was significant for eRT

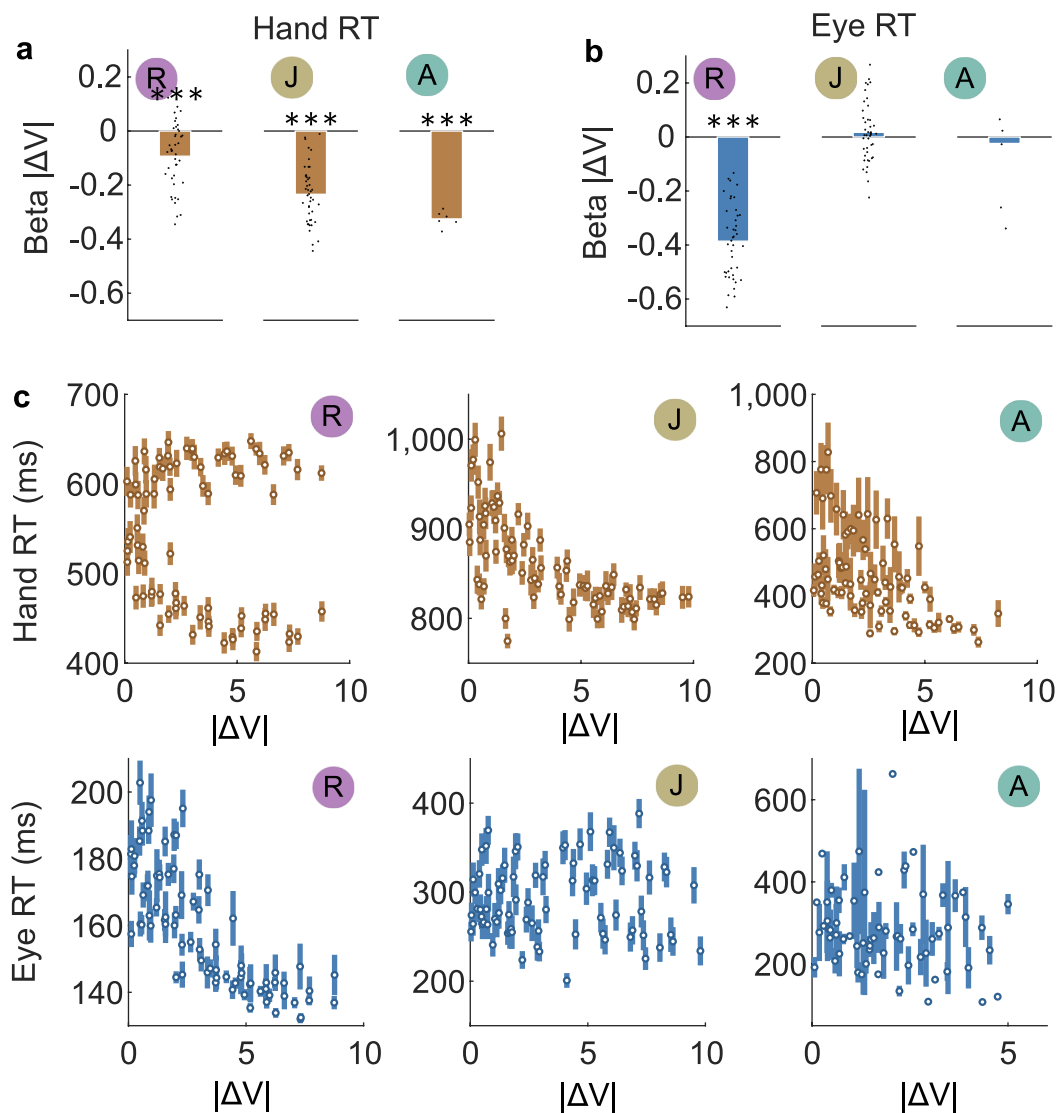


Fig. 6 Sensitivity of reaction time to choice difficulty. **(A)** Result of a linear regression predicting hRT (hand reaction time) from $|\Delta V|$ (subjective choice difficulty): $\text{hRT} = \beta_{\text{cst}} + \beta|\Delta V| \cdot |\Delta V|$. Bars represent the mean value of $\beta|\Delta V|$ and the dots represent the value of $\beta|\Delta V|$ for each session, for each monkey: R, J and A. **(B)** Same as in A but for eRT. **(C)** Dots represent hRT (top) and eRT (bottom) for each of the 81 possible compromises generated by the task, as a function of $|\Delta V|$ (subjective choice difficulty). The vertical bars represent the standard error. * $p < 5 \cdot 10^{-2}$; ** $p < 5 \cdot 10^{-3}$; *** $p < 5 \cdot 10^{-4}$.

($\beta_{\Delta V \times \text{eRT}} = 0.38$, $p = 1.2 \cdot 10^{-15}$) but not for hRT ($\beta_{\Delta V \times \text{hRT}} = -0.07$, $p = 0.26$). For monkey J, the effect was significant in both hRT ($\beta_{\Delta V \times \text{hRT}} = 0.5$, $p = 2.5 \cdot 10^{-12}$) and eRT ($\beta_{\Delta V \times \text{eRT}} = 0.08$, $p = 8.8 \cdot 10^{-9}$). For monkey A, the relation approached significance for hRT ($\beta_{\Delta V \times \text{hRT}} = 0.64$, $p = 0.07$). As expected given the lack of relation between eye movements and the decision in this animal, eRT was not related to $|\Delta V|$ ($\beta_{\Delta V \times \text{eRT}} = -0.03$, $p = 0.87$). Finally, supplementary Fig. 4 shows the same results for the analysis performed without the side-normalization of response time, except for the 'hand' measure of monkey R for which the effect reached significance, only because of the influence of chosen side on reaction time. Thus, after controlling for the effects of $|\Delta V|$ and chosen side on RTs, at least one measure of choice per monkey was more consistent in trials with shorter RTs, which is compatible with the idea that decision accuracy is modulated across trials, potentially by cognitive control.

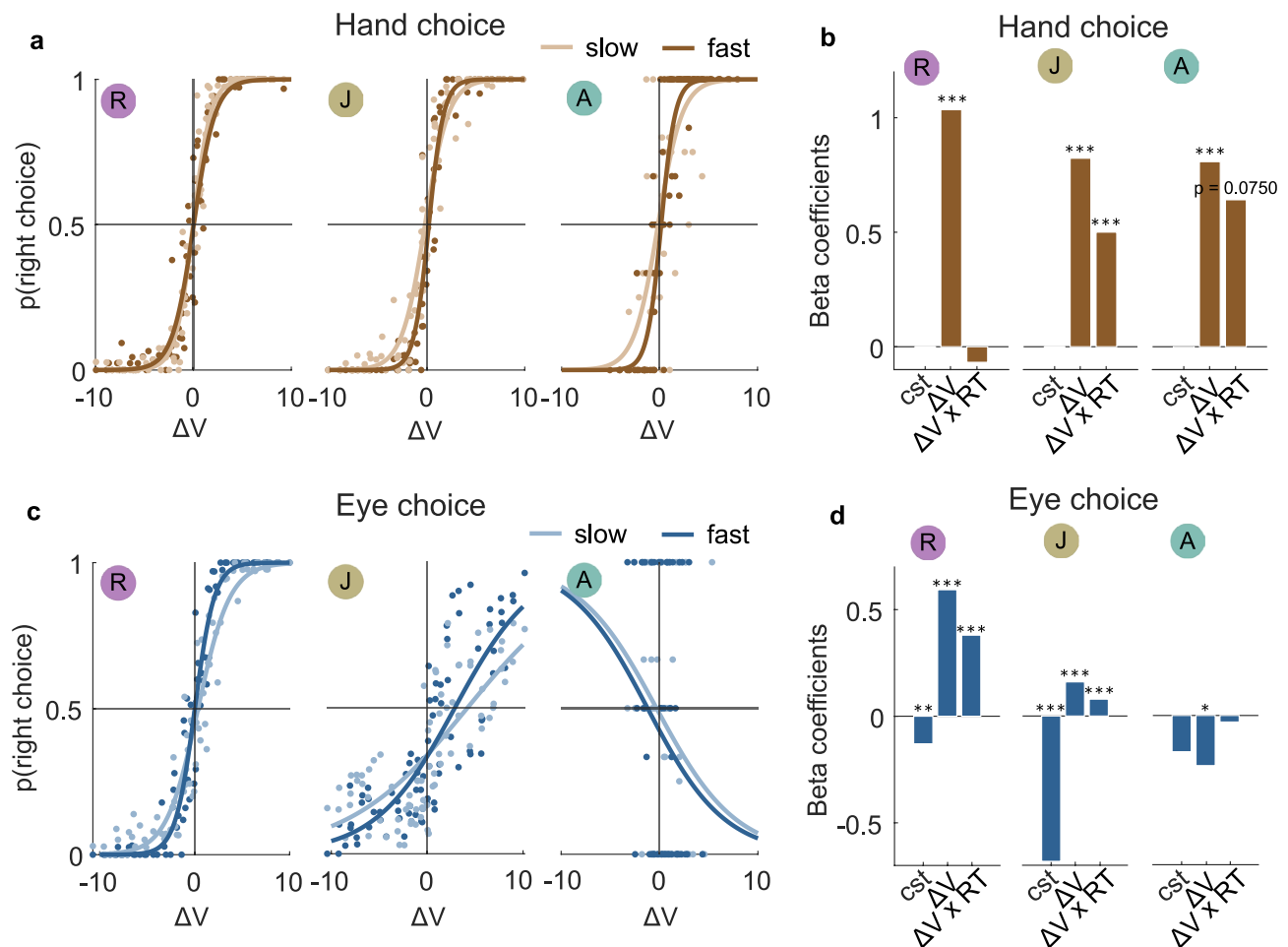


Fig. 7 Correlation between reaction time and choice accuracy. Importantly here, the response times (eRT and hRT) were z-scored within each chosen side prior to the analysis (see “Materials and methods” and Results for further details). The same figure without this transformation can be found in Supplementary materials (Fig. Sup 4). **(A)** Probability to choose the right option depending on the subjective difference of value between the two options (ΔV) on either ‘slow’ or ‘fast’ trials. The slow and fast categories correspond to a separation of all the trials according to hRT (hand reaction time), but with no difference of subjective difficulty ($|\Delta V|$) between the two categories (see “Materials and methods”). The dots represent the proportion of right choices for each of the 81 possible combinations of option, reported to the associated ΔV , during ‘slow’ trials (light orange) or ‘fast’ trial (dark orange). The sigmoid curves are the result of a fit estimating the probability to choose the right option according to ΔV , during ‘slow’ trials (light orange) or ‘fast’ trials (dark orange). **(B)** The bars represent the beta coefficients of a logistic regression that predicts the probability to choose the right option depending on ΔV (right–left) as well as on the interaction between ΔV and hRT: $p(\text{right choice}) = \text{logit}(\beta_{cst} + \beta_{\Delta V} \cdot \Delta V + \beta_{\Delta V \cdot RT} \cdot \Delta V \cdot \text{‘slow’/‘fast’})$. To capture the effect of the interaction between the ΔV and hRT, the regression was performed on all the sessions computed together. The significance of the effect of each parameter is provided directly by the fitglm function used to perform the regression. **C.** This figure presents the same analysis as in A but performed on the eye choice and eRT (eye response time). **D.** The analysis is the same as in B, but for the eye choice. The logistic regression predicted the probability to saccade to the right option depending on the ΔV (right–left) and on the interaction between the ΔV and eRT. * $p < 5 \cdot 10^{-2}$; ** $p < 5 \cdot 10^{-3}$; *** $p < 5 \cdot 10^{-4}$.

Computational models of the control of decision process

To better clarify which variations in the decision process could account for the negative correlation between response time and choice accuracy, we turned to numerical simulations of computational models.

First, we simulated the choice process using a drift diffusion model (DDM). In our DDM simulations, we set the drift rate to the difference between option values and used standard values for the noise magnitude, as well as

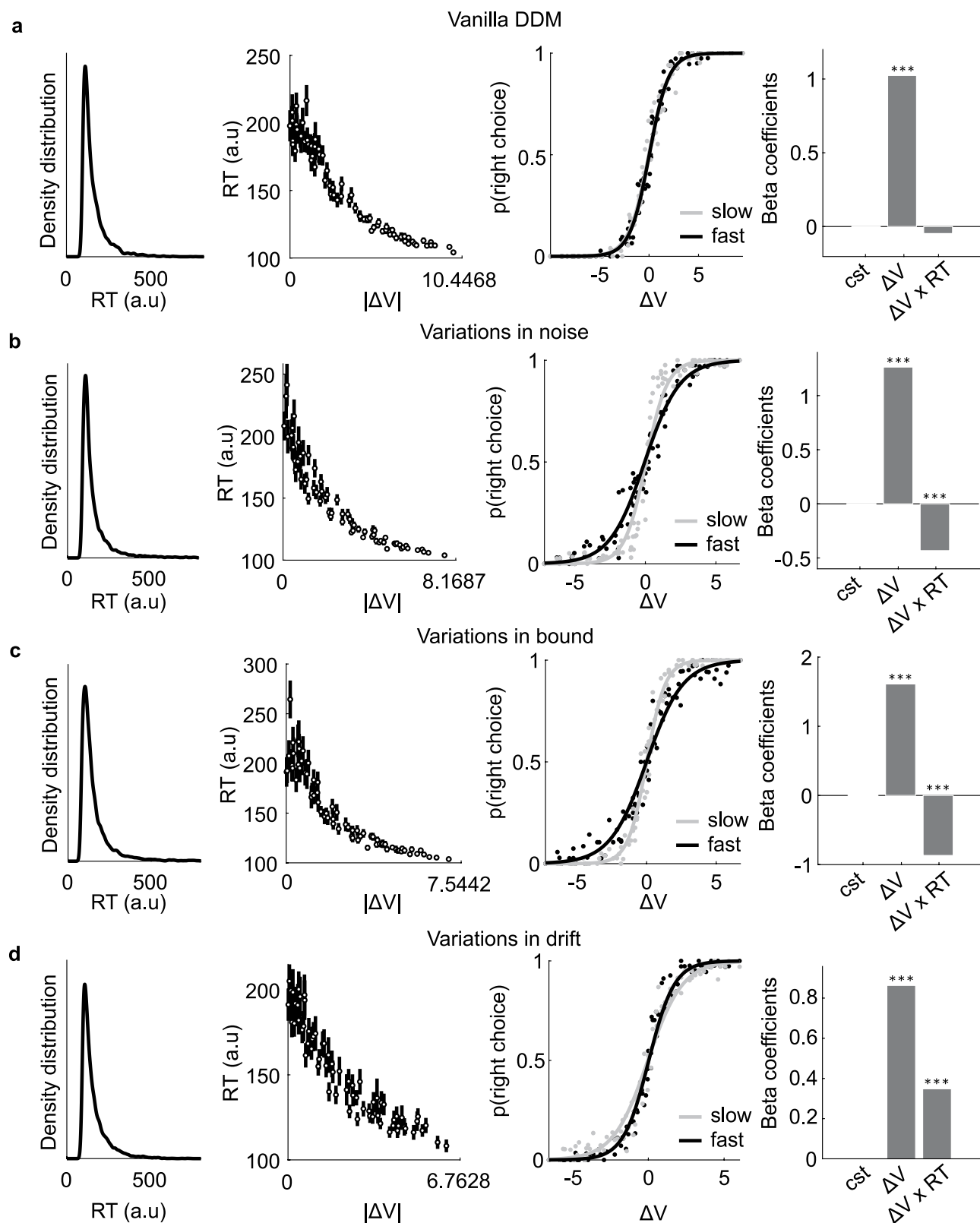


Fig. 8 Drift diffusion model and correlation between reaction time and choice accuracy. Results generated by four different versions of a drift diffusion model (DDM). Each version was generated with standard values for the three principal parameters of the DDM: the drift, the noise and the threshold (see “Materials and methods”). The drift was equal to the ΔV estimated from the choice preferences of monkey R. (A) Vanilla DDM. (B–D) random trial-wise variations ($\mu = 1$, $\sigma = 0.5$) were introduced in either noise parameter (B), bound parameter (C) or drift rate (D) parameter (see “Materials and methods”). The first column presents the distribution density of reaction times expressed in arbitrary unit (a.u.). The second column shows the relation between the reaction time and the choice difficulty ($|\Delta V|$) estimated from the subjective preferences generated by the model (same analysis as in Fig. 5). The third column shows the probability to choose the right option depending on the subjective difference of value between the two options (ΔV) on either ‘slow’ (grey) or ‘fast’ trials (black). This is the same analysis as presented in Fig. 7A, here performed on the response times and choices generated by the models. Choice accuracy is captured by the steepness of the sigmoid function. The fourth column represent the beta coefficients of a logistic regression that predicts the probability to choose the right option depending on the ΔV (right–left) and on the interaction between the ΔV and RT: $p(\text{right choice}) = \text{logit}(\beta_{\text{cst}} + \beta_{\Delta V} \Delta V + \beta_{\Delta V \times \text{RT}} \Delta V \cdot \text{RT} \cdot \text{‘slow/fast’})$. This is the same analysis as presented in Fig. 7C and D but here performed on response times and choices generated by the model. The vanilla DDM (A) cannot generate the negative relation between RT and choice accuracy observed in the data (Fig. 7). Only the model that includes random variations in drift rate (D), can generate that negative relation. Models with random trial-wise fluctuations in noise (B) and boundary levels (C) can only generate a positive relation between RT and choice accuracy. * $p < 5 \cdot 10^{-2}$; ** $p < 5 \cdot 10^{-3}$; *** $p < 5 \cdot 10^{-4}$.

for the upper and lower bounds (see “Materials and methods”). Critically, this simple model does not generate the negative correlation that we observe between response time and choice accuracy (Fig. 8A—right, $\beta_{\Delta V \times \text{RT}} = 0.02$, $p = 0.70$). We thus modified this vanilla DDM to explore which source of trial to trial variability could reproduce the relation between faster response times and greater choice accuracy. We considered three factors: the noise magnitude, the position of the upper bound (decision threshold) and the drift rate. Randomly varying either the noise magnitude or the upper bound across trials induced a positive correlation between response time and choice accuracy (noise: Fig. 8B—right, $\beta_{\Delta V \times \text{RT}} = -0.67$, $p = 8.07 \cdot 10^{-23}$; bounds: Fig. 8C—right, $\beta_{\Delta V \times \text{RT}} = -0.81$, $p = 1.9 \cdot 10^{-29}$). However, randomly modulating the drift rate across trials induced a negative correlation between response time and choice accuracy (Fig. 8D—right, $\beta_{\Delta V \times \text{RT}} = 0.44$, $p = 4.6 \cdot 10^{-13}$). Here, trials with a drift rate higher than average produce faster response time and better choice accuracy than trials with drift rates lower than average. This indicates that the observed relation between RT and choice accuracy in the behavioral data can be better described as trial to trial variations in the efficacy of an evidence accumulation process (i.e., in the DDM drift rate), rather than in a change in the amount of noise or in the decision threshold. Thus, the negative relation between RT and choice accuracy could be accounted for by an arbitrary modulation of evidence accumulation across trials.

To further understand the cognitive processes underlying this relation, we investigated a scenario of metacognitive control where the amount of resources (time) invested in the decision was monitored and adjusted online. The computational model we used, called the online metacognitive control of decision (or oMCD) model, was built upon the assumption that individuals adjust how much cognitive resources to invest in a decision by balancing effort cost against confidence, while adapting to unpredictable variations in confidence gains⁴⁸. It starts from the assumption that investing cognitive resources (control) is costly but improves one’s confidence into the upcoming choice, and it formalizes the policy (regarding when to stop investing resources) that optimally balances resources expenditure and confidence. In this view, a longer response time simply signals a higher duration of cognitive resource investment (or, equivalently: control). The oMCD model include cognitive control efficacy and cognitive control cost parameters (see Methods). In our numerical simulations, we randomly sampled these parameters across agents (but held them constant within agents, across trials). Under this variant of oMCD, the only source of trial-by-trial variations is thus the sequence order in which option features (reward size and required force level) are processed and integrated to eventually yield option values. The model reproduces the tripartite relationship between response time, choice and values that we observe empirically. More precisely, oMCD agents systematically invest more cognitive resources (i.e., longer RTs) when option values are more similar (see Fig. 9A and B and their choices are more consistent with values when less resources are invested, (i.e., shorter RTs; see Fig. 9C and 9d). Importantly, the latter effect emerges irrespective of the apparent choice difficulty (as

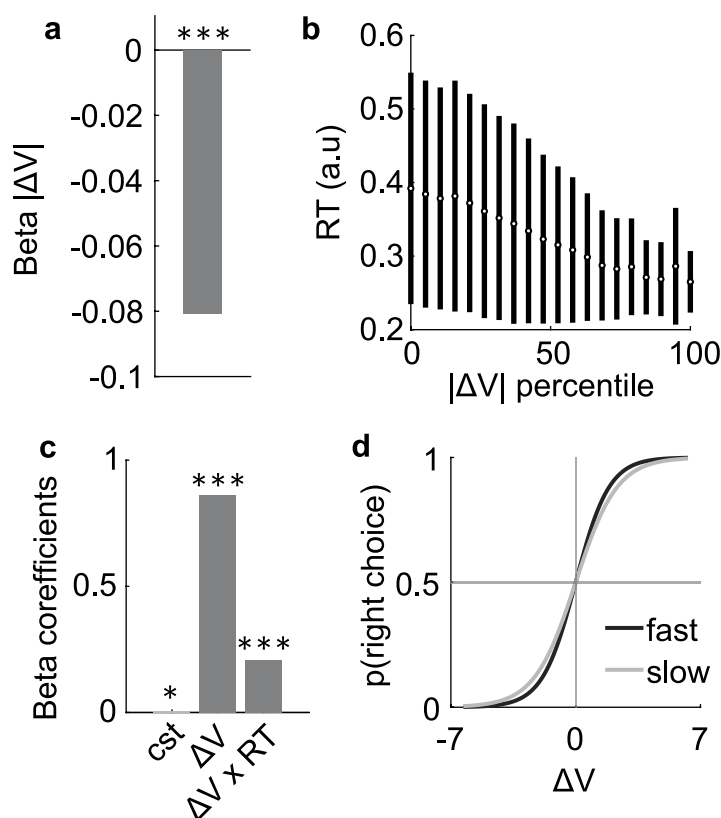


Fig. 9 Tripartite relationship among difficulty, reaction time and consistency under the oMCD model. Results generated by the online metacognitive control of decision (oMCD) model. The model was simulated 50,000 times with random parameters being drawn each time, then each agent generated subjective values, response times and choices in each of the 81 possible compromises of the choice task, without any variation of its parameters. **(A)** Same format as Fig. 6A but using the subjective values and response times generated by the model. For each agent, a linear regression was run across all possible 81 trials to predict the reaction time from the trial difficulty: $RT = \beta_{cst} + \beta_{|\Delta V|} |\Delta V|$. The bar height shows the $\beta_{|\Delta V|}$ averaged across agents, and the vertical bar (invisible on the plot) shows the standard error. As in the real data, the relation between choice difficulty ($|\Delta V|$) and RT is negative. **(B)** Similar format as Fig. 6C but using the subjective values and reaction times generated by the model. For each agent, the distribution of $|\Delta V|$ it could encounter across all possible 81 trials was binned into 20 percentile bins. Each dot shows the RT averaged across agents and trials within a given $|\Delta V|$ bin. Vertical bars show the standard deviation. As in the real data, the relation between choice difficulty ($|\Delta V|$) and RT is negative. **(C)** Same format as Fig. 7B and D, but using the subjective values, response times and choices generated by the model. For each agent, a logistic regression was run to predict the choice probability depending on the subjective value difference and its interaction with the trial category ('slow' or 'fast'): $p(\text{right choice}) = \text{logit}(\beta_{cst} + \beta_{\Delta V} \Delta V + \beta_{\Delta V, RT} \Delta V \cdot \text{slow/fast})$. The bar height shows the three beta coefficients averaged across agents, and the vertical bars (invisible on the plot) show the standard error. **(D)** Similar format as Fig. 7A and C, but using the subjective values, response times and choices generated by the model across agents and trials. Each sigmoid shows the predicted probability to choose the right option separately on slow and fast trials, using a sigmoid $p(\text{right choice}) = \text{sig}(\beta \Delta V)$, where β_{slow} is taken as the coefficient $\beta_{\Delta V}$ averaged across agents, and β_{fast} is taken as the sum of coefficients $\beta_{\Delta V} + \beta_{\Delta V, RT}$ averaged across agents. As in the real data, faster trials are associated with higher accuracy. * $p < 5.10^{-2}$; ** $p < 5.10^{-3}$; *** $p < 5.10^{-4}$.

in monkeys' observed behavior). Indeed, under oMCD, this happens because cognitive control allocation tends to stop before all option features are processed, which render choices made with longer RTs less consistent with the ensuing option values. This model shows how decision relevant resource allocation policies, when optimally trading decision confidence with effort costs, induce the nontrivial relationships between response times, choice and values that we observe in monkey's behavior (both for eye and hand choices).

Discussion

Summary of results

Whether and how rhesus macaques spontaneously control their cognitive efficacy during value-based decisions is unknown. We explored these questions using a binary choice task that involved arbitrating between the two features of each option (reward size and required force level). We first identified a standard signature of cognitive control by assessing the monkeys' ability to overcome counterproductive biases (repetition and side bias). As classically described in situations requiring control, overcoming the bias (in incongruent vs congruent trials) was associated with poorer performance, and might be associated with longer response times. We further examined the potential mobilization of cognitive control by considering another source of choice difficulty: the similarity between the value of the two options. We described the tripartite relationship among choice consistency, choice difficulty and response time. As expected, easier trials were characterized by greater choice accuracy and faster response times. More interestingly, when choice difficulty was held constant, better choice accuracy was still associated with faster response times, which suggests a modulation of choice efficacy beyond mere effects of choice difficulty. In line with the intuition that decision efficacy could benefit from a control process, we also found that choice accuracy was greater in trials where animals displayed a better preparation before cue onset. We then used computational modeling to explore potential mechanisms of decision control, accounting for the joint modulation of choice accuracy and response time. First, we found that standard drift diffusion models could explain monkeys' overt behavior only when assuming that arbitrary trial-by-trial variations of information processing efficacy (drift rate) modulate the impact of difficulty onto choice and response time. Second, we also reproduced the tripartite relation between choice consistency, difficulty and response time with oMCD, which suggests that empirically observed variations in decision efficacy may simply be the outcome of an online supervisory regulation of cognitive resources (using time as a proxy) that optimally trades decision confidence gains with the cost of these resources.

A task to study the control of economic decision

The task was designed to capture variations of monkey's choice efficacy (accuracy) in response to choice difficulty, and evaluate 1) the extent to which the amount of cognitive control invested could regulate the relation between choice efficacy and choice difficulty and 2) how the amount of control invested in the decision could vary as a function of both its expected costs and its expected benefits.

Limitations

We are aware that using decision time as a proxy for the amount of cognitive control invested in the decision might be questioned. First, this is based on the assumption that the amount of control invested for each time unit is constant over the duration of the decision, which is widely accepted in the literature^{16,26,47}. This will be discussed in further details below, when addressing the relation between speed and accuracy. Second, in principle, variations in the duration of the decision process may be influenced by all sorts of decision irrelevant factors^{16,18,64}. However, the task design presents two features that reduce the risk of potential confounds. First, since choices involve a comparison between two options, the influence of individual options in terms of sensory or motor constraint on choice-related RT differences across trials can easily be identified, and excluded from the effects of interest. Second, the separation between engagement epoch and decision epoch allowed us to dissociate the motivational process of deciding whether to engage in the task from the decision process of interest here, i.e., the choice between the two options, as witnessed by the very high rate of trial completion once engaged in the trial (R:98.2%, J: 94.6%, A: 96.8%). Thus, the RT is less likely to be influenced by general (option-independent) motivational effect. Third, the task applied very few constraints on deliberation time (response times rarely

exceed 1500 ms while the time limit was set to 2100 ms). Therefore, the decision process was free to follow its natural course. Thus, we believe that comparing RTs across trials provides a reliable proxy of the contrast in time invested in the decision across the corresponding conditions.

Sources of choice difficulty

The first type of choice difficulty that we examined was the incongruence between the monkeys' biases and the side of their preferred option. We could readily identify two types of bias, a repetition bias and a side bias, which both influenced decisions: in congruent trials, where the bias was favoring the side corresponding to the monkey's preferred option, choices were generally faster and more accurate than in incongruent trials, where monkeys had to overcome the bias to select their favorite option. This is in line with interpretations in terms of cognitive control in classical situation when subjects must overcome a 'default' response to reach a goal^{32,65,66}. Here, by definition, incongruent trials are associated with lower accuracy, because the biases tend to shift the choice away from the monkey's favorite option. The fact that incongruent trials are also associated with increased response times, on average, suggests that this incongruence (between the side of the bias and the side of the monkey's favorite option) constitutes a cognitive challenge, which requires more processing time.

Interestingly, further analysis of the relation among RT, congruence and accuracy indicated that in incongruent conditions, the increase in RT was associated with increased accuracy, which suggests that monkeys could invest time not only because choices were more challenging (because of the incongruence) but also because it might allow them to overcome the bias and improve accuracy, as expected in the general framework of speed-accuracy trade-off^{16,18,26}. Assuming that this investment of time corresponds to an investment of cognitive control, one interpretation of this data is that in incongruent trials, which correspond to a cognitive challenge, monkeys are willing to exert cognitive control in order to maintain performance^{24,49}. But with only two conditions of choice difficulty (congruent vs incongruent), it remains difficult to evaluate quantitatively the mechanisms regulating the allocation of time, which we interpret in terms of cognitive control.

To further quantify the influence of choice difficulty on choice efficacy, we then examined another source of choice difficulty: the contrast in value between the two options. Indeed, choices become increasingly difficult to make as the value contrast between the two options decreases. Note that in all three monkeys, sensitivity to reward size and force level were of similar orders of magnitude, such that decisions involved a relatively balanced arbitration between the two. We could thus generate a wide and homogenous set of choice difficulties, and evaluate the corresponding levels of choice accuracy.

To compensate for the potential noise in the relation between manual RT (hRT), choice difficulty and decision time, we also considered eye-saccades latencies (eRT). We observed that, even if only hand movements were instrumental and eye movements had no influence on the task whatsoever, eye responses were related to hand choices. Indeed, for two of the three monkeys, the direction of the first eye saccade was clearly influenced by the difference in value between the two options, and these monkeys directed their gaze towards the option that they were about to select manually. This is reminiscent of a study by⁶⁷, showing that when subjects were making goal-directed decisions, they directed their gaze towards the chosen option. Note that this pattern was enhanced when subjects were more familiar with the choices and when they had the opportunity to prepare their choice⁶⁷. This is typically the case in our experiment, where monkeys are very familiar with the task and stimuli are not only explicit but also easily identified.

There are other arguments supporting the idea that the two measures were related to a common decision process. Decision making is often considered to involve progressive, internal processing of decision-relevant information, which may not be temporally tied to overt behavioral responses. For example, subjects can make a choice even if no explicit action is required^{68,69} and brain regions involved in the computation of subjective value (here: arbitrating between reward size and force level) are often considered as operating separately from those implicated in triggering a response^{4,70–73}. Eye saccade studies suggest that the actual state of decision is fed continuously to brain regions involved in saccade control, with vigor of saccades reflecting the rise of decision variables

such as confidence and value difference^{74–76}. In addition, value computations are known to be influenced by gaze variations that are otherwise incidental to the decision process^{43,77}. However, the mere initiation of eye saccades can be partially independent from the underlying decision process and could result from an interaction between decision-relevant and decision-irrelevant processes⁷⁸. Altogether, this supports the idea that eRT and hRT provide an indirect, but complementary estimate of the timing of a common decision process. This is in line with previous studies suggesting that eye and limb movements are influenced in a similar way by adjustment related to the relation between speed and accuracy³⁰. For monkey R and J, the differences between the two measures were only quantitative, in that the effects were going in the same direction but showing different magnitudes between the two individuals, presumably because of idiosyncratic differences in the structure of the interactions between the neural systems generating the decision and the ones generating saccades and limb movements. The difference seemed greater for monkey A, in which eye movements did not seem to be related to the decision at all. Altogether, this indicates that, even if the specific dynamics of the decision remain difficult to assess within a trial, it seems reasonable to assume that it took longer on trials where responses times of eyes and/or hand movements were longer, which is sufficient to evaluate a potential effect of covert deliberation time over the decision.

Relation between speed and accuracy

In line with numerous studies, we found that the relation between response time and accuracy did not systematically show the pattern interpreted in terms of speed accuracy trade-off^{16,26,28,30,47,48}. As in those studies, our reasoning remains based on the classical assumption that each time step of the decision contributes to accumulating choice-relevant information, such that, all other things being equal, longer response time should be associated with greater choice accuracy²⁶. Thus, our goal here is to understand the complex regulation of the amount of time invested in the decision, to infer the associated cognitive mechanisms. In that frame, evaluating the relation between choice difficulty (difference in value between the two options), choice efficacy (reliability) and the relative amount of time invested in the choice (RT) can readily provide insight into the influence of control on decision accuracy, given a difficulty level. Based on the two measures of RT in the three monkeys, deliberation time invested in the decision displays a high sensitivity to the subjective value difference, i.e., how easy choices are for the animals: animals responded faster when choices were easier (Fig. 6). Given that difficult choices are also associated with lower accuracy, the increase in RT when the value contrast decreases not only confirms that choices are more challenging in these trials, but also that monkeys are willing to spend more time to make their decision. Thus, as it was the case when considering choice difficulty in terms of congruence between the bias and the options' value, it seems reasonable to interpret this increase in RT in terms of cognitive control allocated to the decision, as usually done in humans^{24,49}.

In line with previous studies in humans and monkeys, we observed a negative relationship between decision time and choice accuracy, when correcting for the effect of choice difficulty (Fig. 7)^{28,30,47,48}. For a given level of choice difficulty, accuracy was greater for shorter RTs, which suggests that the effect of difficulty can be compensated. Note that this apparent compensation is enhanced when monkeys are better prepared before cue onset (Fig. 8). Several intuitions can capture this relation. One possibility is that investing cognitive control leverages more cognitive resources per unit of time, increasing the accuracy of the decision process and resulting in shorter reaction times. Such effect has been observed under the influence of reward incentives in tasks which present a cognitive difficulty, e.g., in the context of conflict processing or task switching^{79,80}. But if the rate of invested cognitive resources per unit of time is fixed, investing more cognitive effort can only increase the duration of decision deliberation, i.e., slowing response time down, as it is the case here in incongruent vs congruent trials^{24,49}. In that case, one potential account of the negative relation between RT and accuracy when controlling for difficulty is that monkeys are not only adjusting the amount of control invested in the decision based on its expected benefits, but also based upon its expected costs, which also scales with RTs^{26,28,47,48}. In other words, monkeys could decide to increase the amount of resources (time) invested in the decision in order to make more accurate choices (and get more favorable outcomes). But they could also decide to invest less resources (i.e.,

respond faster) in the decision because they want to minimize cognitive costs. To better capture the complex and dynamic relation between choice difficulty, choice accuracy and response time, as a proxy for control, we turned to computational modelling.

Using drift diffusion models to capture the relation between choice difficulty, RT and choice accuracy

We started with a simple drift diffusion model, which could capture the relation between RT and choice consistency using simple assumptions^{15,17,19}. Critically, simple versions of the DDM could not fit the data, in line with the intuition that the relation between choice accuracy and RT is modulated by some cognitive processes. Thus, we tried to capture the data by allowing some of the parameters to vary arbitrarily from trial to trial (see Mat. & Meth.). Critically, these different parameters (noise magnitude, drift rate or decision threshold) carry distinct intuitions regarding the source of the trial to trial variability. Here, we showed that only trial to trial variations in drift rate, but not in any of the other two parameters, enable the DDM model to explain the behavioral data of the three monkeys and capture the negative relation between response time and choice accuracy.

How does that translate in cognitive terms? It is well known that the DDM can reproduce many characteristics of the decision process^{15,19}. But what justifies the assumption that DDM's drift rates are varying independently from option values? Recall that the intuition here is that DDM drift rates quantify the efficacy of decision-relevant information processing. However, other modifications of the DDM (e.g., modulations of the noise magnitude and/or the bounds heights) can also predict the opposite effect. This is problematic because all these distinct DDM variants constitute a reasonable implementation of the same idea (e.g., increasing the efficiency of information processing should be equally well modelled by dampening the noise magnitude, or increasing the bounds heights). In other words, under the DDM framework, the impact of variations in processing efficacy may either yield the observed effect or not, depending on modelling strategies that seem a priori arbitrary. Moreover, in that framework, it is more difficult to capture the relation between the time invested in the decision and decision accuracy based on the costs and benefits of that time investment. We addressed this issue using a model which disambiguates the impact of cognitive resource allocation onto decision outcomes: namely, the online metacognitive control of decision—oMCD—model⁴⁸.

Using oMCD to understand why and how control is exerted over the decision

This model frames decision control as a resource allocation problem, where cognitive resources must be invested to process value-relevant information. The idea is that an optimal agent must maximize the probability of making the correct choice (according to its own subjective preferences), while minimizing the cognitive cost of processing value-relevant information. Without any additional assumption, the oMCD model predicts both that higher choice difficulty induces slower and less consistent decisions, and that at constant difficulty levels, higher response time is associated with lower choice accuracy.

Recall that both DDM and oMCD rely on a stopping policy that share a common principle, i.e., a choice is triggered once an internal decision signal reaches a bound. Again, both models are based upon the assumption that investing resources (time) in the decision enhances decision accuracy. The main difference lies in how each framework specifies what the decision signal is, and how the bound is set (for a detailed comparison between the DDM and a former version of oMCD, see¹⁸). In brief, DDM is agnostic about these issues, whereas oMCD commits to a confidence bound that is set in such a way that the ensuing stopping policy maximizes an explicit confidence/cognitive cost trade-off. Although this assumption may appear restrictive, it formalizes simple well established observations. For example, difficult trials are characterized by poor confidence^{55,81,82} and empirical evidence suggests that confidence evolves throughout the decision process⁸³. Note that, under typical DDM variants, it is not clear how the level of accumulated evidence might reflect confidence. This is because most DDM variants assume that the bound is fixed, which implies that all decisions are triggered at the same level of accumulated evidence—and hence, confidence⁸⁴. One way to link evidence accumulation with decision confidence

is to replace the fixed bounds with collapsing bounds^{25,31,85}. Interestingly, some forms of collapsing bounds have been shown to induce stopping policies that optimize some form of speed/accuracy trade-off^{22,25,30,85}. This is essentially because if the demand for evidence decays with decision time, then short decisions will tend to be made with strong confidence. This reasoning, however, does not tell us how to exactly specify the bounds' dynamics. In short, using DDM enables capturing the dynamics of decisions and the investment of control using simple heuristics.

By contrast, oMCD is built to account for a dynamic regulation of control investment. It can formally capture not only the allocation of control based on arbitrary relation between costs and benefits, but also its regulation based on a dynamic evaluation of the cost and benefits of the control itself, in terms of confidence⁴⁸. Under oMCD, bound dynamics directly follow from how the confidence/cognitive cost trade-off is set. Recall that subjective values are internally represented with some uncertainty. As more cognitive resources are invested to assimilate further decision-relevant information, value representations become increasingly certain and choice confidence increases. However, investing cognitive resources implies exerting mental effort, which is inherently costly^{41,86}. As choice confidence increases, expected confidence gains decrease and may thus be overcompensated by the additional cost of investing further resources. Whether confidence is high enough to trigger a decision thus depends on cognitive costs. Interestingly, one can show that the optimal confidence bound may not always obey strictly decaying dynamics⁴⁸. This underlines the importance of laying out assumptions regarding how resource investments determine the way confidence and cognitive costs unfold over time. This also provides a helpful framework for understanding the observed effect of choice difficulty onto decision outcomes. In the early stages of a decision trial, there is ample opportunity for choice confidence to increase. This implies that the optimal policy is to continue unless the current choice confidence is very high. However, as more cognitive resources are invested, expected confidence gains tend to shrink. This will eventually render resource investments not worth the effort, and the optimal policy will shift to stopping even if choice confidence is low. Now, when the decision is difficult, choice confidence increases very slowly. This implies that it will take time for confidence to reach the optimal bound. Difficult choices will thus be slow and achieve low confidence – and hence low accuracy on average.

We clearly observe this predicted relationship in monkeys' behavior. But what is the nature of the underlying resource allocation costs? Note that oMCD does not specify the nature of the invested cognitive resources, which leaves open the issue of what the ensuing cognitive costs are (for a review on the possible sources of cognitive costs, see⁴¹). One possibility is that cognitive (effort) costs arise from the progressive depletion of some limited metabolic resource, which is consumed by the brain systems that operate decision-relevant information processing. Another possibility is that investing limited cognitive resources in a given decision induces opportunity costs, as it involves foregoing the potential benefits of pursuing other goals. In our task, the latency of eye saccades almost never exceeds 500 ms (see Supplementary Fig. 2), which seems very fast for metabolic costs to kick in. One may argue that spending too much time thinking would postpone reward delivery, which may induce some form of delay discounting. In turn, monkeys' behavior may be explained by the imperative to maximize reward rate over time. But slowing down a decision at a given trial would also delay the engagement in the next trial, thus inducing some form of opportunity costs across trials. Discriminating between these explanations would require modifying the current experimental design, e.g., by fixing the delay between cue onset and reward delivery (such that variations in decision time may not induce variations in delay discounting).

This model also provides insight into the sources of trial-by-trial variability in decision outcomes. More precisely: how can choice confidence and RT differ between trials that are similar in terms of option features? Just as the DDM does not specify where the noise of evidence accumulation comes from, the oMCD model is agnostic regarding the origin of the fluctuations of choice confidence accumulation. In fact, the nature – and properties – of this variability lies in the companion “value synthesis” system that processes relevant information. Here, we assumed that option values are modified by progressively integrating features or attributes (namely: reward size and required force levels). In this view, arbitrary variations in the order in which option attributes are considered lead to variations in the evolution of value estimates, and consequently, in the evolution of choice confidence.

Note that this scenario is unrelated to arbitrary variations in mental preparation (prior to cue onset), which seems to bear on choice accuracy above and beyond the effect of choice difficulty. Whether the latter source of trial-by-trial variations is bound to a similar form of metacognitive control, or whether it reflects mere modulations in the monkeys' motivational state or simply neural noise is still to be determined.

Finally, oMCD provides a critical insight into inter-individual variability. Indeed, in the present experiment, the 3 monkeys showed very distinct sensitivity to effort and reward, as reflected in the different sensitivity for the distinct behavioral responses (hand response, eye responses, effect of preparation). Based on model-free measures, it could have been concluded that they were using qualitatively distinct strategies to perform the task. Based on model-based analysis, however, it appears that the differences across the 3 monkeys are only quantitative. Indeed, even if their sensitivity to cost and benefits differ, all 3 monkeys seemed to use the same strategy (oMCD) to adjust the level of cognitive control they want to invest in the decision, based on a trade-off between the cost of this control and the expected benefits in terms of choice accuracy.

Implications for comparative cognition

These results provide a new insight into the cognitive mechanisms underlying decision making in primates. Importantly, our goal here was not to establish whether rhesus macaques could perform tasks that justify the use of psychological concepts such as cognitive control or metacognition to describe their behavior. Rather, we wanted to identify the nature of the computational processes underlying their choices. We found that, as humans, rhesus macaques can rely upon a complex modulation of decision efficacy, using a confidence-based, supervisory evaluation of the expected costs and benefits of the cognitive resources allocated to the decision. In humans, based on well accepted operational definitions, this is interpreted as an online metacognitive control decision accuracy, where the level of control allocated to the decision is itself supervised by a metacognitive (confidence based) process⁴⁸. Is it legitimate to apply the same reasoning in monkeys? A vast number of studies have demonstrated the existence of several forms of cognitive control in monkeys and apes^{50–52}. Several studies have also demonstrated the existence of metacognitive monitoring for actions and memory^{57,59,60,87}. Thus, given that the computational process involved in the decision (oMCD) appears similar between humans (Benon et al., 2024) and monkeys (present study), it seems legitimate to refer to this computational process as a metacognitive regulation of control allocation. Indeed, this idea has been proposed before for rhesus macaques performing perceptual tasks^{61,63,88}. But again, our conclusions regarding the nature of the cognitive operations underlying choices in macaques are based on the operational properties of the computational model, rather than on their ability to perform any diagnostic tasks. As such, the label used to refer to these computational features does not affect the conclusion.

Conclusion

In line with the literature, this study shows that rhesus macaques can trade speed for accuracy when incentivized to do so, which can be interpreted in terms of allocation of cognitive control in the decision. Critically, the complex dynamic relation among choice accuracy, choice difficulty and control allocation can be captured by considering not only the benefits of time (control) investment, but also its cost, as it is the case in humans. This provides a clear demonstration that economic decisions can be controlled and monitored in non-human primates. Further comparative studies, including other species, would enable a better understanding of the evolution of these cognitive processes in primates.

Materials and methods

Ethical statement

Experimental procedures were designed in association with veterinarians of the ICM Paris Brain Institute, approved by the Regional Ethical Committee for Animal Experiment (CREEA IDF no. 3), and performed in compliance with the European Community Council Directives (2010/63/UE) at the ICM (Agreement number: A-75-13-19).

Animals

Three male adult rhesus monkeys (Monkey R: 10 kg, Monkey J: 17 kg, Monkey A: 19 kg) were used for this experiment. All animals had been used in previous behavioral experiments and were habituated to the lab environment and task setup. To control their motivation in the task, the animals' access to water was restricted during periods of behavioral testing. On testing days, they had unlimited access to fluid rewards through task performance. However, if the amount of water obtained during the task was not sufficient to fulfill their physiological needs, they received water supplementation after the session. They had free access to water whenever testing was interrupted for more than a week. In brief, whether it was a testing day (in which they received water from the task) or not, they received enough water to fulfill their physiological needs on a daily basis. Monkeys also had free access to food.

All 3 monkeys had been implanted with a head fixation post and they were always head fixed while performing the task, to allow for the monitoring of eye movements (see Varazzani et al., 2014 for further details on the procedure). Sessions usually lasted approximately one hour, and never more than two hours. Monkeys only completed one session per day.

Behavioral task

The behavioral task was designed and monitored using the EventIDE software (OkazoLab, London, United Kingdom). Monkeys were squatting in a restraining chair (Crist Instrument Co, Hagerstown, MD, USA) in front of a computer screen which displayed the visual cues related to the task. Three touch sensitive bars (M2E, Eau-bonne, France), (left, middle and right) were installed on the chair at the level of monkey's hands. The monkeys received water from a tube placed between their lips.

Monkeys performed a binary choice task, where each of the 2 options was characterized by a trade-off between a given amount of fluid reward and a given level of force demand, which corresponded to the amount of force that monkeys had to exert on a grip to obtain the reward (Fig. 1). At each trial, Reward and force levels of each of the 2 options were indicated by visual cues. Monkeys selected one of the 2 options (left vs right) by squeezing the corresponding grip, and the expected reward was delivered if animals exerted the required force. Force and reward levels (1–3) were generated randomly and independently from each other, for each of the 2 options. Thus, there were 9 possible combinations per option (Reward size x Force level), resulting in 81 possible combinations of the two options).

The force levels were calibrated independently for each of the 3 monkeys and could be adapted depending on side (left or right). The first force level was set just above the baseline, such that a minimal force would be sufficient to obtain the reward. The third force level corresponded to the maximum force that monkeys could exert, as determined using an incremental approach during pilot calibration sessions. The second level was intermediate between the other two. The three reward sizes corresponded to 3, 4 and 7 drops of liquid (diluted grenadine or apple juice).

Each trial started with the onset of a red dot at the center of the screen (Fig. 1). After a short period (100 ± 50 ms) the red dot turned green (go signal), after which the animal had 6000 ± 200 ms to touch the middle bar, and

thereby engage in the decision task. Touching the bar was indicated by visual feedback (the central dot turned blue), which stayed on for 900 ± 100 ms before the cues appeared. Then the cues indicating reward size and force level, along with a green dot, were displayed on each side of the screen. The required force level was signaled by the size of a vertical bar and the reward size was signaled by the number of drops (Fig. 1). The animal had up to 2100 ms to evaluate the options before selecting one, by pressing the corresponding lateral bar (left or right). As soon as the monkey reached the required force level, the corresponding dot turned blue (feedback) and the reward was delivered. If for any reason monkeys failed to complete the trial (omission or failure to reach the required force level), the trial was marked as erroneous and a new trial started from the initiation period with the same options.

Data analysis

Analysis of the data was performed using the MATLAB software (MATLAB 2020a, Mathworks).

The mean number of performed trials per session was: 169 ± 7 SEM, 159 ± 6 SEM and 89 ± 15 SEM for monkey R, J and A respectively. The number of sessions included in the analysis was 71, 47 and 9 for monkey R, J and A respectively. Eye-tracking data was obtained for 41, 44 and 6 sessions for monkey R, J and A respectively.

We analyzed five trial-wise behavioral variables. Two of them were measuring choice preferences (binomial variables): direction of the first saccade (see Eye-tracking analysis) and chosen side. We also computed two measures of response time (RT, continuous variable): an ‘eye’ RT (eRT) and a ‘hand’ RT (hRT). The eRT was defined as the delay between the cue onset and the first saccade (see Eye-tracking analysis). The hRT was defined as the delay between cue onset and the beginning of the force exertion on the selected lever (left or right). Finally, we also measured trial-wise success in performing the action as a binary variable, (1 when the exerted force was above the required force level, 0 otherwise).

Analysis of economic choice

We computed the subjective value of each option, as defined by the weighed sum of the conflicting option features (i.e., reward size and force level). To do so, we performed a logistic regression of choice preferences onto option features, using the data from all sessions:

$$\text{Choice} \sim \text{logit}(c + \beta_{\text{forceR}} \cdot \text{forceR} + \beta_{\text{rewardR}} \cdot \text{rewardR} + \beta_{\text{forceL}} \cdot \text{forceL} + \beta_{\text{rewardL}} \cdot \text{rewardL}) \quad (1)$$

where ‘Choice’ (binomial variable) is the choice revealed by the instrumental action (i.e., side bar press: right = 1, left = 0), c is a constant (lateralization bias) term, forceR and forceL are the required force level for the right option and left option, respectively, and rewardR and rewardL are the reward size for the right and left option, respectively. The corresponding β weights are unknown sensitivity parameters. The logit function gives the probability to choose the right option given the features of both options, therefore each β weight’s sign and amplitude quantifies how each option feature was beneficial (positive sign) or detrimental (negative sign) for choosing the right option. Therefore, the value of each option can be defined as:

$$V_{\text{right}} = c + \beta_{\text{forceR}} \cdot \text{forceR} + \beta_{\text{rewardR}} \cdot \text{reward} \quad (2)$$

$$V_{\text{left}} = -\beta_{\text{forceL}} \cdot \text{forceL} - \beta_{\text{rewardL}} \cdot \text{reward} \quad (3)$$

We defined the subjective value difference ΔV as:

$$\Delta V = V_{\text{right}} - V_{\text{left}} \quad (4)$$

Note that a choice is consistent with the underlying value difference if the monkey prefers the right (left, respectively) option when $\Delta V > 0$ ($\Delta V < 0$, respectively). The above logistic regression adjusts the unknown β weight parameters such that they maximize choice consistency (though this leaves some inconsistent trials, see below).

Analysis of decision control to overcome bias

We also quantify the influence of two potential bias on the choice: a side bias, and a repetition bias. We used the following logistic regression where ΔR and ΔF are respectively the differences in reward size and required force level between the two options, and prev.choice is the choice at the previous trial (left vs right).

$$\text{Choice} \sim \text{logit}(c + \beta_{\Delta R} \cdot \Delta R + \beta_{\Delta F} \cdot \Delta F + \beta_{\text{prev.choice}} \cdot \text{prev.choice}) \quad (5)$$

The side bias is captured by the constant (c) and the repetition bias is captured by the weight of the previous choice ($\beta_{\text{prev.choice}}$).

To assess whether overcome these bias to choose the best option in regards to reward and force could constitute a cognitive difficulty, we analyzed the relation between the congruence between side indicated by the bias and side indicated by the best option. For this we first computed the subjective difference in value between the two options ΔVal , which diverse from the previously introduced ΔV in that it does not take into account the side bias.

$$\Delta \text{Val} = \beta_{\Delta R} \cdot \Delta R + \beta_{\Delta F} \cdot \Delta F \quad (6)$$

Indeed, both ΔV and ΔVal were computed using ΔR and ΔF , but ΔV also included the constant.

Then, we used a logistic regression to predict choice accuracy (congruence between the best side according to ΔVal and the chosen side), according to the congruence between the best side and the side indicated by the bias, and the absolute difference of value between the two options $|\Delta \text{Val}|$ which had an expected effect on choice accuracy.

$$\text{Accuracy} \sim \text{logit}(c + \beta_{\text{congru}} \cdot \text{congruence} + \beta_{\text{adv}} \cdot |\Delta \text{Val}|) \quad (7)$$

We repeated this analysis two times to assess the effect of the side bias, and the repetition bias.

Finally, to assess whether the cognitive difficulty of overcoming the bias could be countered by mobilization of cognitive resources, using hRT as a proxy, we used a model to predict choice accuracy according to $|\Delta \text{Val}|$, the congruence between the best side and the side indicated by the bias, hRT and the interaction between hRT and the congruence.

$$\text{Accuracy} \sim \text{logit}(c + \beta_{\text{adv}} \cdot |\Delta \text{Val}| + \beta_{\text{congru}} \cdot \text{congruence} + \beta_{\text{RT}} \cdot \text{hRT} + \beta_{\text{inter}} \cdot \text{congruence} \times \text{RT}) \quad (8)$$

Again, this analysis was repeated two times, for the side bias and the repetition bias.

Analysis of the relation between accuracy, choice difficulty and reaction time.

We also studied with more details the relation between RT and choice performance. As RTs were strongly influenced by choice difficulty (which also influenced the consistency between choice and the underlying value difference ΔV), we decided to categorize the trials into ‘fast’ and ‘slow’ trials for each possible level of choice difficulty. More precisely, we examined all sessions together, and for each of the 81 possible choice combinations, we did a median split on the RTs. The trials in which the RT was above median were labelled as ‘slow’ and trials in which the RT was below median were labelled as ‘fast’. We captured the influence of RT on sensitivity to ΔV by introducing the interaction between ΔV and RT in a model of choice already taking into account the influence of ΔV (see Eq. (4)).

$$\text{Choice} \sim \text{logit}(c + \beta_{\Delta V} \cdot \Delta V + \beta_{\Delta V, \text{slow/fast}} \cdot \Delta V \cdot \text{slow/fast}) \quad (9)$$

where the 'slow/fast' variable encodes the trial category (1: slow trial, -1: fast trial), and $\beta_{\Delta V, \text{'slowfast'}}$ captures the influence of RT on ΔV sensitivity.

Binarization of RT also allowed us to plot the regression curves associated with either fast or slow RT for the sake of illustration of the effects.

One possible confounding factor was the correlation between RT and chosen side, which was significant for some measures. Thus, we also performed this analysis with a corrected version of RT. That is, we normalized RT from chosen side by z-scoring separately the RT in right choice trials and RT in left choice trials. This effectively removes the main effect of chosen side from RT data. We performed this analysis on the 2 types of measure: direction of first saccade in relation to its RT (eRT) and chosen side in relation to its RT (hRT).

Drift diffusion model

To capture the relation between response time and choice consistency, we compared different Drift Diffusion Models -DDM- variants. Briefly, the DDM is a stochastic model of choice and RT generation, which relies on the assumption that a decision is triggered once some accumulated evidence has reached a predefined threshold. Unknown DDM parameters include: evidence drift and noise magnitudes, as well as two evidence thresholds (upper and lower bounds, i.e., one for each option). To assess how the DDM could account for different directions of correlation between RT and choice consistency (none, positive or negative) independently from choice difficulty, we simulated a vanilla DDM (Model1) as well as 3 other DDM variants in which either the drift, the noise magnitude or the thresholds were randomly modulated over trials:

$$\text{Model1 : } V_{t+1} = V_t + \delta + v_t; \beta_1 = 500, \beta_2 = 0; V_0 = \beta_1/2$$

$$\text{Model2 : } V_{t+1} = V_t + \alpha \cdot \delta + v_t; \beta_1 = 500, \beta_2 = 0; V_0 = \beta_1/2$$

$$\text{Model3 : } V_{t+1} = V_t + \delta + \alpha \cdot v_t; \beta_1 = 500, \beta_2 = 0; V_0 = \beta_1/2$$

$$\text{Model4 : } V_{t+1} = V_t + \delta + v_t; \beta_1 = \alpha \cdot 500, \beta_2 = 0; V_0 = \beta_1/2$$

Here, V_t is the value of evidence accumulation at time t (after cue onset), δ is a drift parameter which takes the value of ΔV at each trial, and v_t is a time-varying gaussian noise variable centered on 0 (with standard deviation $\sigma=22$). Regarding DDM parameters: β_1 and β_2 correspond to the value of upper and lower bound respectively, V_0 is the value of evidence accumulation at time $t=0$ (this parameter may be used to simulate a side bias), and α is a random parameter the value of which was set at each trial, following a gaussian distribution centered on one ($\mu=1$, $\sigma=0.5$). We also implemented a time of non-decision during the first 100 time steps of each trial, during which δ and v_t terms are equal to 0.

We simulated choices using those models on 6000 trials (equivalent to the number of trials in our experimental data for monkeys R and J). The ΔV (i.e., drift parameter, δ) was determined using the estimated sensitivity to force level and reward size for monkey R (Eq. (1) and (2)). Then we performed the same analysis as described above (Eq. (1), (2), (3), (4) and (9)) to determine the relation between simulated RT and simulated choice consistency, for each DDM variant.

Online MCD model

To better understand the mechanistic link between mental effort and the processes at play during decision making, we then simulated choices under the oMCD model⁴⁸.

Briefly, the model assumes that the uncertain value V_i estimated for each considered option is represented as a normal distribution with mean μ_i and variance σ_i , where $i \in \{1,2\}$ designates the index of the option in a binary choice task:

$$V_i \sim N(\mu_i, \sigma_i) \quad (10)$$

These value estimates are updated as cognitive resources are leveraged to process decision relevant information. In the following, $\mu_i(z)$ and $\sigma_i(z)$ respectively designate the mean and variance of the value estimated for option after having invested an amount of resources z . We also note $\Delta\mu(z) = \mu_1(z) - \mu_2(z)$.

Note that the model assumes that the option with the highest value mean is always chosen, without any additional noise in the choice policy. Here, possible errors arise from the fact that value estimates might be wrong (e.g., because the choice is triggered before all option features have been processed – see below), which can only be known after the consequences of the choice are realized.

Nevertheless, choice confidence $P_c(z)$ at any within-decision time can be evaluated, in terms of the probability that the option with the highest mean value is indeed the best option. Such decision confidence simply measures how discriminable are the value distributions, which is directly determined by the moments of the two value distributions $\mu_1(z)$, $\mu_2(z)$, $\sigma_1(z)$ and $\sigma_2(z)$. We refer to the article from⁴⁸ for more details regarding the mathematical derivation of choice confidence.

Importantly, the metacognitive controller cannot know in which direction the means of value estimates will shift before additional information is processed. However, one can show that the following agnostic assumptions regarding how first-order and second-order moments of the value distribution change as resource investment unfolds are sufficient to enable quasi-optimal decision control:

$$\frac{1}{\sigma_i(z + \delta z)} = \frac{1}{\sigma_i(z)} + \beta \quad (11)$$

$$\begin{cases} \mu_i(z + \delta z) = \mu_i(z) + \delta\mu \\ \delta\mu \sim N(0, \gamma) \end{cases} \quad (12)$$

Here, β is called the *type #1 effort efficacy* and controls the rate at which the precision of value estimates increases, and γ is called the *type #2 effort efficacy* and controls the magnitude of the future changes in the mean value estimates.

Finally, investing cognitive resources z entails a cognitive cost:

$$C(z) = \alpha z^\nu \quad (13)$$

where α is called the *unitary effort cost* and ν is called the *cost power*.

The oMCD model then consists of maximizing the expected balance between these costs and choice confidence. The optimal amount of resources to invest $\hat{z}$ is:

$$\hat{z} = \underset{z}{\operatorname{argmax}} P_c(z) - C(z) \quad (14)$$

This is achieved by comparing, at each time point, the current trade-off $P_c(z) - C(z)$ with the best expected trade-off if more resources are invested. The agent stops if the current trade-off exceeds any future trade-off and continues investing resources otherwise.

To simulate dynamical value estimates and predict choices between specific pairs of options, the oMCD model must be combined with a “value synthesis” system, i.e., a scenario that specifies how values are actually updated over time. Here, since the task involves options defined through two attributes (force level and reward size), we used an attribute integration model, assuming that the true subjective value V_i^* of an option is computed from the weighted sum of its attributes:

$$V_i^* = \omega_R R + \omega_F F \quad (15)$$

where R and F designates the reward size and force level associated with option $i \in \{1, 2\}$, which are respectively weighted by the parameters ω_R and ω_F . Note that ω_F is typically negative, as the required force level is an aversive

attribute. We also work under the assumption that monkeys process information attribute per attribute. When the attribute of an option has not been processed yet, the value synthesis system should rely upon its sample mean (i.e., the mean reward size or force level, across trials) instead.

Of course, this model of value synthesis generates distributions of estimated values which are not normal, and the dynamics of value synthesis might not follow exactly those considered by the oMCD model (Eq. 10 and 3). However, it is possible to set the β and γ parameters to approximate within-trial dynamics of choice confidence under this value synthesis scenario (see⁴⁸ for details).

We then simulated 1000 agents with various subjective preferences, cost function, and prior biases, all uniformly distributed: for each agent, we randomly selected ω_R between 0.5 and 1.5, ω_F between -1.5 and -0.5, α between 0.1 and 0.5, ν between 0 and 4, and the prior mean value difference $\Delta\mu(0)$ between -1 and 1. We then derived the approximated effort efficacy parameters β and γ from the weights ω_R and ω_F . Finally, we exposed each agent to all 81 possible choices combinations (as in the experiment). For each choice, we considered all possible permutation orders of processing of option attributes (since there are four attributes in total, there are 24 possible orders). Each sequence of attributes, defined as a given set of four attributes processed in a given order, produced a unique dynamics of choice confidence, which updates throughout a trial as more and more attributes are integrated to estimate values. These dynamics defined both a reaction time (the moment when the optimal policy is to stop investing resources) and a choice (the option with the highest mean estimated value at the stopping time). For each choice combination, reaction time and choices were then averaged across all possible sequences to derive the mean reaction time and the probability to choose the left option.

Force signal analysis

The force signal was digitized and sampled at 1 kHz. Reaction time to reach lateral bar (hRT) was computed by detecting the moment when the signal started to rise continuously during 20 ms and exceeded a threshold corresponding to 8 standard deviations of the signal at rest (signal before cues onset). Note that modifying these criteria did not alter the qualitative nature of the results.

Eye-tracking analysis

We used eye-tracking measures to assess the relation between eye-movement and choice process in our task. Monkeys were positioned approximately 1 m away from the monitor. Eye position was collected using an eye-tracking system (ETL 200, ISCAN, Ma, USA). Eye position was acquired at 240 Hz. Within experimental sessions, focal distance and magnification were kept constant after calibration. We recorded the position of the pupil of the right eye on 2 axes (lateral and vertical) at 1 kHz. Calibration consisted in identifying the position of the eye when monkeys fixated the center of the monitor, where the fixation spot initially appeared. As cues were separated along the lateral axis, only the lateral position of the pupil was analyzed. Before analysis, we used a simple pre-processing algorithm to detect blinks (aberrant and extreme changes in signal) and correct the signal by interpolating missing values with a theoretical monotonous function spanning from prior to posterior aberrant values.

A preliminary analysis (Supplementary Fig. 1) confirmed our observations during data acquisition: monkeys tended to fixate the center of the screen during trial initiation, and they tended to look at the chosen option around the time of the choice. In order to quantify these processes, we assessed the timing of the triggering of the first saccade towards one of the 2 sides after the onset of the cues. Here, saccades were detected by putting a threshold corresponding to 4 standard deviations on the temporal derivative of the signal.

For the analysis of saccades after cue onset, we quantified the dynamics of the first saccade occurring after cue onset. We measured the reaction time and the direction of the first saccade occurring at least 100 ms after cue onset, which is assumed to be the time necessary for visual information to reach higher order areas of the cortex^{89,90} and for visual cues to trigger decision related saccades⁹¹.

Statistics

General linear model analyses were performed using the internal `fitglm` function of MATLAB. For each linear model, continuous variables were always z-scored. When variable sampling was sufficient at the level of the session we tested the significance of the regression weights with a test at the second level: the weights (betas) were estimated for each behavioral session and a t-test was performed to assess if the mean was different from zero (across sessions). This approach was favored to confirm that effects of interest were stable across sessions. Otherwise, we performed the glm after computation of all sessions together and gave the p-values of each weight provided by the `fitglm` function of MATLAB. Mean comparisons were performed using appropriate statistical tests after assessing for normality and homogeneity of variance. Eventual proportion comparisons were performed using a z-test. Eventual correlations between variables were assessed using internal `corrcoef` MATLAB function.

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Data availability The data are currently stored on the ICM server, and they are readily available upon request.

Declarations

Competing interests The authors declare no competing interests.

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